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[Intervention Review]

Clinical effectiveness of telepharmacy services in patients with non-communicable diseases in ambulatory care settings

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ABSTRACT

Rationale

Non-communicable diseases (NCDs), such as hypertension, diabetes, and asthma, require continuous medication management. However, medication adherence remains suboptimal. Telepharmacy—defined as pharmacist-led care delivered remotely via telephone, video, or digital platforms—may improve adherence and clinical outcomes while addressing access barriers, but uncertainty remains regarding clinical effectiveness and generalisability. A systematic review is warranted to assess whether telepharmacy improves medication adherence, safety, and other key outcomes compared with usual care.

Objectives

To assess the clinical effectiveness of telepharmacy services, compared with usual care, on medication adherence and clinical outcomes in patients with NCDs in ambulatory care settings.

Search methods

We searched CENTRAL, MEDLINE, Embase, Global Index Medicus, and two trial registries up to 15 December 2025. We also assessed the reference lists of included studies and relevant reviews, conducted citation searching, and contacted study authors to clarify information and identify additional data. No language or publication status restrictions were applied.

Eligibility criteria

We included individually randomised controlled trials (RCTs) and cluster-RCTs comparing pharmacist-led telepharmacy with usual care for people with NCDs (e.g. cardiovascular disease, diabetes, and cancer) in ambulatory care settings.

Outcomes

Critical outcomes were medication adherence, patients' satisfaction, and drug-related problems (DRPs). Important outcomes included mortality rate, worsening of NCDs, clinical measurements, laboratory values, patients' quality of life, healthcare use, and economic outcomes. We included seven outcomes in the summary of findings table.

Risk of bias

We assessed the risk of bias for the seven outcomes in the summary of findings table using the Cochrane RoB 2 tool, incorporating both individually RCTs and cluster-RCTs.

Synthesis methods

We conducted synthesis analyses using random-effects models, calculating summary risk ratios or mean differences (MDs)/standardised mean differences (SMDs) with 95% confidence intervals (CIs). For cluster-RCTs, we used adjusted estimates or applied design effect corrections. Where meta-analysis was not feasible, we used narrative synthesis. We assessed the certainty of the evidence using GRADE.

Included studies

We included 21 trials (17 individually RCTs and 4 cluster-RCTs) involving a total of 5440 participants with NCDs. Sample sizes ranged from 20 to 1400 participants. Studies were conducted in high-, upper-middle-, and lower-middle-income countries, across hospital, clinic, pharmacy, or insurer-based settings. Interventions targeted conditions such as diabetes, hypertension, and asthma. Telepharmacy interventions varied in delivery modes (e.g. telephone, video, and app), intensity, and components (e.g. adherence support, monitoring, and education). Follow-up durations ranged from one to 18 months, with most studies lasting 12 months or less.

Synthesis of results

Telepharmacy interventions may improve medication adherence compared with usual care (SMD 0.32, 95% CI 0.10 to 0.55; 10 studies, 2978 participants; low-certainty evidence).

For patients' satisfaction, the evidence is very uncertain about the effect of telepharmacy interventions compared with usual care (SMD 0.37, 95% CI -0.11 to 0.85; 3 studies, 422 participants; very low-certainty evidence). One additional study using a 5-point Likert scale reported little to no difference between groups (96.5% versus 97.5%; $P = 0.68$). Another study lacked a comparator group, and we excluded it from the synthesis.

We did not pool the evidence for DRPs due to clinical and methodological heterogeneity. Narrative findings from individual studies showed that one study reported increased detection of DRPs. Other studies reported fewer adverse events, suggesting prevention of DRPs, while the remaining studies found no clear differences. The certainty of the evidence was low.

Regarding important outcomes, two studies reported worsening of NCDs. Due to clinical heterogeneity, we did not pool the results and presented them narratively. The effect of telepharmacy on worsening of NCDs remains uncertain.

For asthma control, no clear difference was observed (SMD 0.23, 95% CI -0.34 to 0.80; 2 studies, 318 participants). Telepharmacy interventions may reduce systolic blood pressure (SBP) (MD -6.82 mmHg, 95% CI -12.16 to -1.48; 5 studies, 1254 participants; low-certainty evidence) and may reduce diastolic blood pressure (DBP) (MD -2.50 mmHg, 95% CI -4.80 to -0.20; 5 studies, 1254 participants; low-certainty evidence) compared to usual care. Two additional studies reporting clinical measurements found more pain relief with telepharmacy in one study, and no clear difference in thromboembolic events in the other.

For glycated haemoglobin (HbA1c), telepharmacy interventions probably have little or no effect (MD -0.10%, 95% CI -0.25 to 0.05; 5 studies, 1771 participants; moderate-certainty evidence). For LDL cholesterol, a meta-analysis of two studies showed no clear difference between the groups (MD -0.84 mg/dL, 95% CI -4.70 to 3.02; 2 studies, 444 participants). One study reported better prothrombin time-international normalised ratio (INR) control in the intervention group.

Three studies assessed quality of life using different tools, but did not show consistent evidence of benefit. Hospital admissions and emergency department visits showed no clear differences between groups.

Two studies evaluated economic outcomes, with one reporting cost savings and the other showing no difference in total or disease-related costs.

No included studies reported data on mortality rate or adverse events attributable to telepharmacy, so potential harms remain uncertain.

Authors' conclusions

Low-certainty evidence suggests that telepharmacy interventions may improve medication adherence, and may reduce both SBP and DBP in patients with NCDs in ambulatory care settings compared to usual care. Moderate-certainty evidence indicates telepharmacy interventions probably have little or no effect on HbA1c. The evidence is very uncertain about the effect of telepharmacy interventions on patients' satisfaction. The evidence base is limited by short follow-up periods, variation in interventions and outcome measures, and lack of equity-related data. Telepharmacy appears promising for ambulatory care, but further high-quality trials with standardised adherence measures and longer follow-up are needed to clarify effectiveness, implementation potential, and equity impacts.

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Registration

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PLAIN LANGUAGE SUMMARY

Does telepharmacy (remote pharmacy care) work better than usual care for people with long-term health conditions who are not in hospital?

Key messages

- Telepharmacy may help people take their medicines as prescribed, and may reduce both systolic blood pressure (the pressure in your arteries when your heart beats) and diastolic blood pressure (the pressure between heartbeats). It probably makes little or no difference to blood sugar control (HbA1c).
- We are very uncertain about its effects on patients' satisfaction with care. Telepharmacy may affect medicine-related problems (such as side effects or interactions), but findings varied because some interventions aimed to detect more problems while others aimed to reduce harms from medicines. No studies reported the number of deaths or unwanted effects from telepharmacy, so potential harms remain uncertain.
- Studies did not show consistent effects on quality of life, hospital admissions, hospital emergency department visits, or healthcare costs. Most studies involved only a few people or had limitations, and the studies varied in delivery and content.

What is telepharmacy?

Telepharmacy is the use of phone calls, video calls, mobile apps, or other electronic communication to provide pharmacy services without meeting in person. It can include advice about medicines, help with taking medicines as prescribed, and monitoring of health. Telepharmacy can make it easier for people to get support from a pharmacist, especially if they live far from a clinic or have difficulty travelling. It may also help pharmacists detect medicine-related problems and adjust treatments when needed.

What are long-term health conditions?

Long-term health conditions, sometimes called chronic conditions, are health problems that can usually be controlled but not cured. They last for three months or longer, and may get worse over time. People with long-term conditions often require ongoing care. Examples include diabetes, high blood pressure, asthma, and heart disease. These conditions can affect a person's quality of life and may lead to serious complications if not well managed. People with long-term conditions often need regular check-ups, tests, and support to take their medicines correctly.

What did we want to find out?

We wanted to find out whether telepharmacy, compared with usual care, helps people take their medicines as prescribed, and improves patients' satisfaction with care, medicine-related problems, asthma control, blood pressure, and blood sugar control (measured using the HbA1c test) in people with long-term conditions in outpatient settings. We also looked at other possible effects, such as quality of life, admissions to hospital, emergency department visits, and healthcare costs.

What did we do?

We searched for studies that compared pharmacist-led telepharmacy with usual care for people with long-term conditions in outpatient settings. We included high-quality studies, known as randomised controlled trials. We summarised the results and rated our confidence in the evidence, considering factors such as study design, number of participants, and consistency of results.

What did we find?

We found 21 trials including 5440 people, which were performed in different countries and settings (e.g. pharmacy, hospital). Most trials compared telepharmacy with usual face-to-face care for people with long-term conditions in outpatient settings. In the included studies, usual care generally referred to standard face-to-face care without additional telepharmacy support. Most studies lasted 12 months or less.

In people with long-term conditions in outpatient settings, compared with usual care:

- telepharmacy may improve how well people take their medicines as prescribed (10 studies, 2978 people), that is, as instructed by their healthcare providers;
- the evidence is very uncertain about whether telepharmacy improves patients' satisfaction with their care (3 studies, 422 people);
- telepharmacy may affect problems such as medicine side effects or interactions (5 studies, 547 people), but findings varied because some studies aimed to detect more problems while other studies aimed to reduce harms;
- telepharmacy may reduce systolic blood pressure (the pressure in your arteries when your heart beats) and diastolic blood pressure (the pressure between heartbeats) (5 studies, 1254 people); but
- telepharmacy probably makes little or no difference to blood sugar control (HbA1c levels) (5 studies, 1771 people).

No studies reported the number of deaths or unwanted effects from telepharmacy, so potential harms remain uncertain. We have more confidence in some findings (such as blood sugar control) than in others (such as patients' satisfaction with their care). For some outcomes, future research could change what we know.

What are the limitations of the evidence?

Our confidence in the findings was reduced because many studies involved only a few people, had problems with how they were carried out, or produced results that did not always agree. Evidence about problems caused by medicines came only from descriptions given by patients, as we were unable to group and analyse data from different studies. The studies used telepharmacy interventions that varied in their content, and how or how often they were given, which may have influenced the results.

How up to date is this evidence?

The evidence is up to date to December 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Telepharmacy interventions compared to usual care for patients with non-communicable diseases

Telepharmacy interventions compared to usual care for patients with non-communicable diseases

Patient or population: patients with non-communicable diseases

Setting: ambulatory care settings

Intervention: telepharmacy interventions

Comparison: usual care

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with usual care	Risk with telepharmacy interventions				
Medication adherence assessed with: the Morisky Medication Adherence Scale, 8-item (MMAS-8), as a representative measure Scale from: 0 to 8 follow-up: range 3 months to 12 months ^{a,b}	-	SMD 0.32 SD higher (0.1 higher to 0.55 higher)	-	2978 (10 RCTs)	⊕⊕⊕⊕ Low ^{c,d}	Telepharmacy interventions may improve medication adherence compared to usual care.
Patients' satisfaction assessed with: the Telepharmacy Metered-Dose Inhaler (MDI) Technique Evaluation instrument, as a representative measure Scale from: 1 to 5 follow-up: range 8 days to 12 months ^e	-	SMD 0.37 higher (0.11 lower to 0.85 higher)	-	422 (3 RCTs)	⊕⊕⊕⊕ Very low ^{f,g}	The evidence is very uncertain about the effect of telepharmacy interventions on patients' satisfaction.
Drug-related problems (DRPs) follow-up: range 8 days to 12 months	Findings from five studies varied and reflected differences in outcome constructs. Detection: one study (Ibrahim 2022), which aimed to improve detection of DRPs, reported a higher number of DRPs identified in the intervention group. Prevention: four studies aimed to reduce adverse drug events and generally reported numerically fewer events in the intervention group. One additional study (Salmany 2018) described adverse effects in 20% of participants but did not pro-			547 (5 RCTs)	⊕⊕⊕⊕ Low ^{h,i,j}	-

	vide group-level data, so it could not contribute to the analysis. The results were not pooled due to heterogeneity in outcome definitions and intervention intent. Only studies contributing to this narrative synthesis were considered when summarising participant numbers in the summary of findings table.					
Mortality rate - not reported	-	-	-	-	-	No included studies reported mortality outcomes.
Clinical measurements (SBP) assessed with: SBP follow-up: range 6 months to 12 months	The mean clinical measurements (SBP) was 134 mmHg	MD 6.82 mmHg lower (12.16 lower to 1.48 lower)	-	1254 (5 RCTs)	⊕⊕⊕⊕ Low ^{k,l}	Telepharmacy interventions may reduce SBP compared to usual care.
Clinical measurements (DBP) assessed with: DBP follow-up: range 6 months to 12 months	The mean clinical measurements (DBP) was 81 mmHg	MD 2.5 mmHg lower (4.8 lower to 0.2 lower)	-	1254 (5 RCTs)	⊕⊕⊕⊕ Low ^{k,m}	Telepharmacy interventions may reduce DBP compared to usual care.
Laboratory values (glycated haemoglobin [HbA1c]) assessed with: HbA1c follow-up: range 3 months to 12 months	The mean laboratory values (glycated haemoglobin [HbA1c]) was 8.51 %	MD 0.1 % lower (0.25 lower to 0.05 higher)	-	1771 (5 RCTs)	⊕⊕⊕⊕ Moderate ⁿ	Telepharmacy interventions probably have little or no effect on HbA1c compared to usual care.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_459669255493338367.

- ^a Abbreviations: DBP: diastolic blood pressure; DRPs: drug-related problems; GRADE: Grading of Recommendations Assessment, Development and Evaluation; HbA1c: glycated haemoglobin; MDI: metered-dose inhaler; MMAS-8: 8-item Morisky Medication Adherence Scale; RCT: randomised controlled trial; SBP: systolic blood pressure.
- ^b Medication adherence was assessed using both validated self-reported scales (e.g. MMAS-8; Medication Adherence Report Scale [MARS]) and objective measures derived from pharmacy refill records (e.g. medication possession ratio [MPR]; proportion of days covered [PDC]).
- ^c Downgraded by one level because 10 of 11 studies were at high risk of bias, while one study was at low risk.
- ^d Downgraded by one level due to substantial heterogeneity ($I^2 > 75\%$), and the direction of effect was not consistent across studies (two studies favoured usual care).
- ^e Patients' satisfaction was assessed using various instruments, including the Telepharmacy Metered-Dose Inhaler (MDI) Technique Evaluation instrument, a Visual Analogue Scale (VAS), and an overall healthcare satisfaction rating.
- ^f Downgraded by two levels because all four studies were at high overall risk of bias.
- ^g Downgraded by one level due to serious imprecision. The confidence interval included both no effect and a meaningful benefit, and the total sample size was small (3 studies, 211 participants), not meeting the optimal information size.
- ^h Downgraded by one level because 4 of 5 studies were at high risk of bias, while one study was at low risk.
- ⁱ Not downgraded for inconsistency. Although one study (Ibrahim 2022) aimed to increase detection, the others aimed to reduce adverse events and showed consistent effects, with no differences between groups.
- ^j Downgraded by one level due to imprecision. Most studies had small sample sizes and reported non-significant or borderline effects. Findings relied on narrative synthesis, without pooled effect estimates, and the optimal information size was not met.
- ^k Downgraded by one level due to overall risk of bias, with some concerns across all five included studies.
- ^l Downgraded by one level due to considerable heterogeneity ($I^2 = 88\%$) and variation in the direction and size of effects.
- ^m Downgraded by one level due to substantial heterogeneity ($I^2 = 73\%$), indicating inconsistency in the direction and magnitude of effects across studies.
- ⁿ Downgraded by one level due to two studies at high risk of bias and two with some concerns, although one study contributing the largest weight was at low risk.

BACKGROUND

Description of the condition

Management of chronic diseases is one of the greatest challenges for healthcare systems worldwide [1]. Chronic diseases are defined as conditions that can usually be controlled but not cured, last for three months or longer, and may get worse over time [2]. Chronic conditions can be categorised as non-communicable diseases (NCDs), persistent communicable diseases (e.g. HIV infection), long-term mental health disorders (e.g. depression and schizophrenia), or ongoing impairments in structure (e.g. amputations, blindness, and joint disorders) [1]. NCDs include mainly cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes. NCDs usually require long-term treatment [3, 4].

The World Health Organization's (WHO) nine voluntary global targets of NCDs are as follows:

1. 25% relative reduction in overall mortality from cardiovascular diseases, cancer, diabetes, or chronic respiratory diseases;
2. at least 10% relative reduction in the harmful use of alcohol;
3. 10% relative reduction in prevalence of insufficient physical activity;
4. 30% relative reduction in mean population intake of salt/sodium;
5. 30% relative reduction in prevalence of current tobacco use in persons aged 15 years or older;
6. 25% relative reduction in the prevalence of raised blood pressure, according to national circumstances;
7. halt the rise in diabetes and obesity;
8. at least 50% of eligible people receive drug therapy and counselling to prevent heart attacks and strokes; and
9. 80% availability of affordable basic technologies and essential medicines [5].

These can be achieved by ensuring proper treatment coverage and access to basic technologies and medicines [5]. In 2021, the 10 leading causes of death were responsible for 39 million deaths, accounting for 57% of the 68 million deaths reported worldwide [6]. NCDs represented seven of the 10 leading causes of death and accounted for 38% of all deaths globally [6]. Major NCD-related causes of death included ischaemic heart disease, stroke, chronic obstructive pulmonary disease, trachea, bronchus and lung cancers, Alzheimers disease and other dementias, and diabetes mellitus [6]. According to the WHO, age-standardised NCD mortality rates in low-, lower-middle-, and upper-middle-income countries were 640.4, 589, and 496 per 100,000, respectively, compared to 338.1 in high-income countries (HICs) in 2019 [7]. The global risk of death from any of the four major NCDs for adults (between the ages of 30 to 70 years) in low- and middle-income countries (LMICs) was double that of HICs in 2019 [7]. These findings are consistent with the WHO 2021 update, in which NCDs continued to account for a substantial proportion of global mortality and several major NCDs remained among the leading causes of death worldwide [6, 7]. The WHO has highlighted the importance of implementing treatment and prevention strategies for NCDs to reduce disease progression and healthcare burden globally [5].

The treatment and prevention of further worsening of NCDs often requires continuous medication management. In 2003, the WHO

reported the medication adherence rate to long-term treatment for chronic diseases was only around 50% in HICs, and the rates were even lower in LMICs [8]. Data from 2014 reported for patients with NCDs in the USA showed high rates of medication non-adherence to NCD medications: 76% non-adherence with one medication, and 32% non-adherence with two or more medications [9]. In contrast, the low and moderate adherence rate among hypertensive patients in the study population in 12 LMICs was 30.8% and 33.6%, respectively [10]. Furthermore, the pooled proportion of non-adherence to anti-diabetic medications in five studies was 43.4% using an 80% adherence cut-off [11]. These data suggest that medication adherence, irrespective of socioeconomic setting, is still not optimal.

Therapeutic drugs are often prescribed, administered, and used inappropriately. Successful and safe drug therapy requires individualised drug therapy decisions for each patient, the achievement of a concordance (i.e. an agreement between the patient and the healthcare professional about the outcome of treatment and how it will be achieved), and patient monitoring [12]. Medication errors and inappropriate prescribing are recognised as major problems for healthcare systems, both clinically and economically [13]. For people living with NCDs, most healthcare encounters and medication use therefore occur outside hospital admission, with patients spending substantially more time receiving care in outpatient and community settings than as inpatients. In this context, ambulatory care refers to health services provided to patients who are not admitted to hospital and do not require overnight stays, and encompasses a broad range of outpatients, community-based, home-based, and remote care services [14]. Within ambulatory care settings, pharmacists may contribute to patient care through a range of activities, including medication review and optimisation, patient education and counselling, support for medication adherence, monitoring of treatment outcomes, identification and management of drug-related problems (DRPs), and collaboration with other healthcare professionals [15, 16].

Description of the intervention and how it might work

Role of pharmacists

The role of pharmacists has evolved substantially over time. Until the 1960s, pharmacists' responsibility was mainly to review prescriptions and dispense medications [17]. Since then, this healthcare field has been presented with many challenges: increasing health demands combined with the complexity of the usage of medicines and patients' poor medication adherence, and advancements in technology. These challenges in turn have created more opportunities for pharmacists and expanded their scope of practice [12, 17, 18, 19]. Pharmacists now provide patient counselling not only on medications but also on lifestyle modifications, medication therapy management (MTM), and disease state management in various practice settings. Research shows that pharmacists have contributed positively to improved medication adherence and patient clinical outcomes through various interventions, such as face-to-face patient counselling, remote or telephone-based interventions, and medication review for chronic diseases, such as asthma, hypertension, hyperlipidaemia (high cholesterol), diabetes, and cardiovascular diseases [15, 16, 20, 21]. However, the pharmacists' interventions reported in these studies were predominantly face-to-face [15, 16, 21].

Digital health interventions

Broadly, eHealth is defined as the use of information and communication technologies to improve public health reporting, promote disease surveillance, and facilitate disease monitoring [22]. Within this broad framework, digital health interventions (DHIs) represent a category of eHealth and include mobile and other digital technologies [23]. DHIs aim to support healthcare systems through various functions, including the provision of healthcare services between patients and providers at a distance, remote consultations, monitoring of patient health or diagnostic data, transmission of medical information, and case management consultations between healthcare providers [23, 24]. These functions are delivered through digital approaches, such as telehealth and telemedicine. Telehealth refers to both non-clinical services (e.g. video conferencing and transmission of still images) and clinical services delivered at a distance, whereas telemedicine is limited to clinical services [25].

Telepharmacy

Telepharmacy is defined as an approach used in pharmacy practice, where a pharmacist utilises telecommunication technology to facilitate and provide patient care services at a distance [26, 27]. It can include activities such as counselling sessions via telephone, remote reminders, counselling through telecommunications technology such as video, or multifaceted interventions. There is some evidence of telepharmacy services improving medication adherence rates and targeted clinical outcomes [26, 27]. Telepharmacy may be particularly beneficial in supporting geographically isolated communities with medication use when there are an insufficient number of pharmacists or other healthcare providers [28]. Various programmes have been conducted in which pharmacists effectively used telephone and video conferencing technology to provide counselling to outpatients with chronic disease and HIV infections in remote areas [29, 30, 31]. The provision of remote MTM services for diabetes or hypertension patients, or both, in rural areas has also resulted in improved health and safety outcomes while reducing potential healthcare disparities [32]. In Denmark, a chat-based telepharmacy has been running nationally since 2012, dealing with a variety of enquiries, including DRPs, symptoms, and more technical questions [33]. For outpatients, remote video conferencing and telephone communication with pharmacists increased satisfaction and improved access to drug therapy and pharmacy services [29, 34, 35].

In addition, telepharmacy services have also been associated with improved clinical measurements, such as glycosylated haemoglobin (HbA1c) and blood pressure [36]. A meta-analysis of anticoagulation services using telepharmacy in the outpatient setting found that the telepharmacy service was effective [37]. The telepharmacy service was associated with a lower risk of bleeding and hospitalisation compared with face-to-face services [37]. Telepharmacy has also been used in acute care to strengthen the support system between hospitals. For instance, as part of tele-intensive care unit (ICU) services, telepharmacy consultation that focused on drug-drug interaction review and dosage adjustment in patients with both renal and liver failure had improved medication safety and reduced DRPs [38]. A systematic review also observed the benefits of telepharmacy in an acute care setting [39]. It showed that the communication between pharmacists and other healthcare providers through telephones and video conferencing

supplemented the usual pharmacist services and improved nurses' satisfaction [39].

Chronic diseases, such as NCDs, often require a long period of supervision, observation, or care [40]. The Chronic Care Model (CCM), developed in the 1990s, is a framework for improving patient health outcomes through a multifaceted and organised approach to healthcare systems [41]. The model is designed to work for individual patients, the community of patients and providers, and the healthcare system [41]. In addition, a number of other strategies exist to improve the quality and comprehensiveness of chronic disease management in the community [42, 43]. In providing patient-centred care, these models have assisted with the implementation of continuity of care and prompt access to care [42]. Though the management of NCDs in ambulatory care plays an important role in both the prevention and treatment of NCDs [40], it is necessary to understand which interventions are effective, for whom, and in what circumstances, in order to improve care.

The benefit of telepharmacy is to facilitate access to health care, and it can reduce the cost of care. Telepharmacy is beneficial in urban areas and has substantial advantages for rural populations, such as faster access to appropriate professional expertise, or avoidance of movement on the part of patients and healthcare providers [24, 44]. Telepharmacy improves medication use in communities with a limited number of pharmacists. An example of such communities is geographically isolated ambulatory care clinics and healthcare facilities [28]. In LMICs and other high- and middle-income countries, pharmacists providing non-dispensing services in-person have been shown to improve clinical outcomes and quality of life, such as blood pressure, blood glucose, and cholesterol levels, albeit to a lesser extent, and to reduce healthcare costs and health service utilisation related with rates of hospitalisation and emergency department (ED) visits [16]. However, even in high-income settings, there is a gap between urban and rural areas in terms of access to medication [35]. Several studies reported that telepharmacy enabled the provision of more comprehensive pharmacist-led services compared with usual care, particularly in settings with limited access to in-person pharmacy services [45, 46, 47, 48]. There is a need for healthcare systems that leverage limited human and financial resources effectively while maximising the contribution of relationships between patients, populations, and healthcare providers to health outcomes. Telepharmacy may represent one approach to improving the quality and efficiency of healthcare delivery, particularly in geographically extended or resource-constrained settings.

In the implementation of telepharmacy, there are different ways of intervening and communicating, not only with patients and populations but also between healthcare providers [37, 39, 49, 50]. A systematic review examining the effects of telepharmacy on anticoagulant management considered the following study and intervention characteristics when evaluating intervention effects across studies: mode of communication, type of intervention, country of publication, provider characteristics, and comparator services [37]. The review showed that telepharmacy was as effective as face-to-face patient/healthcare provider interaction [37]. In addition, few studies reported benefits from real-time consultations compared with asynchronous ones. Some studies observed that ongoing consultations were more beneficial compared to one-time consultations [51]. The implementation of

pharmacist-led interventions through telepharmacy services might be useful in optimising the quality and efficiency of pharmaceutical care in the long term in large geographical regions, and could be an effective way to address the gaps in access to healthcare.

Why it is important to do this review

Several systematic reviews have examined the effects of telepharmacy on health outcomes and other factors [37, 39, 49, 50]. In communities worldwide, research shows that pharmacists have helped improve patient care by providing management services, mostly through direct face-to-face service, for NCDs such as cardiovascular diseases, hypertension, diabetes, and asthma [15, 16, 21]. However, evidence on the impact of pharmacists' led interventions provided via telepharmacy on clinical outcomes in ambulatory care settings is still limited, and much still needs to be studied to help guide decision-makers [37, 49, 50].

A published systematic review of telepharmacy was conducted in both ambulatory care and acute care settings, and included populations with chronic diseases and certain treatments such as anticoagulant management [51]. Totten and colleagues published a non-Cochrane systematic review that involved a narrative review of 34 studies testing the effect of clinical pharmacist telemedicine interventions on clinical outcomes, such as clinical disease management, in outpatient or ambulatory settings [51]. The review focused on overall chronic disease management in adults, including NCDs (e.g. hypertension, diabetes), long-term mental diseases (depression), persistent communicable diseases (e.g. HIV), and smoking cessation. It included randomised trials and also retrospective cohort studies. The review authors did not search the Cochrane Central Register of Controlled Trials (CENTRAL). The review was reported as a narrative of an overall positive effect on outcomes related to clinical disease management in the management of chronic diseases without quantitative analysis and assessing the risk of bias [51]. Additionally, the studies identified did not systemically assess the benefits of video communication over telephone or other intervention methods. Given the diversity of patients' access to health care, it is also necessary to examine intervention methods.

Since this review was conducted in 2016 by Niznik and colleagues [50], at least 10 randomised trials have been published that are potentially eligible for inclusion in our Cochrane review. We plan to conduct a quantitative analysis of the clinical effectiveness of pharmacist-led interventions by restricting it to randomised trials for NCDs. The 2018 systematic review in the acute care setting by Strnad and colleagues assessed the effect of telepharmacy services for critically ill inpatients at ICU and non-ICU [39]. Although the review demonstrated a positive effect on patient outcomes of a feasible intervention as narrative, it was not an intervention for chronic patient conditions [39]. A 2020 systematic review comparing the impact of different types of telemedicine on anticoagulation control and adverse outcomes in patients taking warfarin included pharmacists, nurses, pharmacy technicians, and automated management system telemedicine intervention providers [49]. An automated management system with telephone and web-based follow-up is an approach and possibility for optimal anticoagulation management by patients themselves. In addition, the use of telemedicine is comparable to usual care, so it may be necessary to consider it as an alternative to face-to-face warfarin management where appropriate in reducing both barriers to access and costs, and to investigate whether incorporating interaction

with healthcare providers could reduce the number of adverse outcomes [49].

One systematic review and meta-analysis assessed the effectiveness of pharmacist-led anticoagulation management by telepharmacy in outpatients. Telepharmacy was comparable to face-to-face services and conventional physician care in terms of clinical outcomes, such as thromboembolic and major bleeding events and hospital admissions, with fewer hospital admissions in the telepharmacy group [37]. The effectiveness of telepharmacy has been reported in quantitative systematic reviews focusing on anticoagulants for the treatment and prevention of antithrombotic events in cardiovascular disease in the ambulatory care setting. No previous systematic review has identified a quantitative analysis of the clinical effectiveness of telepharmacy that included an outpatient population with other NCDs, such as cancers, chronic respiratory diseases, and diabetes [37, 49]. In ambulatory care settings, there is a need to assess the clinical effectiveness of telepharmacy services in treating and preventing further deterioration of chronic conditions, such as NCDs, which require longer care. The use of telepharmacy services has expanded substantially, especially since the COVID-19 pandemic [52]. However, much still needs to be assessed to help guide decision-makers [52].

OBJECTIVES

To assess the clinical effectiveness of telepharmacy services, compared with usual care, on medication adherence and clinical outcomes in patients with NCDs in ambulatory care settings.

METHODS

Differences between protocol and review

In the Cochrane protocol [53], we categorised the outcomes as primary and secondary. At the review stage, we reclassified outcomes as either critical or important, in line with current Cochrane and GRADE guidance [54, 55], to reflect their relevance for evidence certainty assessment and summary of findings tables. This change represents a terminological and reporting update, and did not involve changes to the outcomes assessed.

We included medication adherence as a critical outcome in this review. In the Cochrane protocol [53], we did not pre-specify a particular measurement tool for adherence. Therefore, we included all measures of medication adherence as reported by the study authors, including self-reported questionnaires (e.g. 8-item Morisky Medication Adherence Scale (MMAS-8)), pill counts, and pharmacy refill data [56].

At the protocol stage, we described some outcome categories broadly (e.g. clinical measurements, laboratory values, and healthcare utilisation). At the review stage, we disaggregated these for more appropriate analysis. Specifically, we divided clinical measurements into asthma control, systolic blood pressure (SBP) and diastolic blood pressure (DBP), pain control, and thromboembolic events; laboratory values of HbA1c, low-density lipoprotein (LDL) cholesterol, and prothrombin time-international normalised ratio (PT-INR); and healthcare utilisation into unplanned hospital admissions and ED visits.

We used Covidence to manage screening and data extraction [57], which we did not specify in the protocol.

We clarified the description of the risk of bias assessment in the [Methods](#) section by explicitly stating that we assessed the risk of bias at the outcome level for all outcomes prespecified in the protocol [53], including the relevant outcome measures and time points. This clarification did not represent a change to the planned methods.

We added a post hoc sensitivity analysis for medication adherence, excluding one study that used an unvalidated outcome measure, to assess the robustness of the findings.

In addition, we revised the outcomes included in the summary of findings table. Composite outcomes for clinical and laboratory values were disaggregated based on data availability and outcome characteristics. Specifically, we separated clinical measurements into SBP and DBP, and we separated laboratory values to include HbA1c. Due to space limitations of the summary of findings table (maximum of seven outcomes), we did not include all prespecified outcomes. Through these changes, we aimed to improve interpretability.

Criteria for considering studies for this review

Types of studies

We included individual randomised controlled trials (RCTs) and cluster-RCTs. Cluster-RCTs were included to avoid the risk of contamination of interventions among patients in the same facility, as we expected the intervention method of this review to include unblinded trials. We excluded trials with a cross-over design to avoid potential carry-over effects. Also, we excluded non-randomised trials, including quasi-RCTs. Controlled before-and-after studies, interrupted time series, and repeated measures studies were excluded. We included studies published as full-text articles, abstracts only, and unpublished data, regardless of publication status or language.

Types of participants

We targeted participants/populations with NCDs (e.g. cardiovascular diseases, cancers, and diabetes). We examined how pharmaceutical care via telepharmacy addressed critical and important outcomes for NCDs. All remote interventions led by pharmacists, such as patient education, counselling, and monitoring services in any clinical setting (e.g. hospital, community pharmacy, nursing home, residential care) were included, and the care setting focused on patients in ambulatory care rather than inpatients receiving medical treatment in the hospital. We did not set exclusion criteria based on sex, location, ethnicity, or socioeconomic status. Also, we included remote pharmacy care for children.

Types of interventions

We included trials comparing telepharmacy services with usual care. We focused on telepharmacy services that provided remote patient counselling, education, and monitoring, and clinical services by pharmacists. We defined a remote intervention as one that utilised telecommunications technology instead of traditional face-to-face pharmacy operations or patient care services. Telecommunication technologies included synchronous video (two-way interaction), asynchronous content (stored audio or video), remote patient monitoring, or mobile health (mobile devices; telephone, email, and short message/messaging service (SMS)) [27].

The comparisons for this review were usual patient care, in which pharmacists provided services only via direct face-to-face patient treatment. We limited telepharmacy services to those in which patients received remote advice from pharmacists. We excluded interventions provided to patients via another healthcare professional. In addition, we excluded trials in which telemedicine was performed by multiple professionals because it might be difficult to ascertain the effectiveness of telepharmacy only.

Outcome measures

We assessed a range of outcomes, including medication adherence (regardless of assessment method used, including self-reported questionnaires, pill counts, and pharmacy refill data), patients' satisfaction, DRPs, laboratory values for each disease, healthcare process outcomes (such as the number of unplanned hospital admissions and hospital visits), and economic outcomes. We included adverse events caused by telepharmacy and drug treatment. The outcome measures in trials did not constitute part of the inclusion/exclusion criteria.

Critical outcomes

1. Medication adherence
2. Patients' satisfaction
3. DRPs, such as adverse drug events (ADEs) and inadequate drug use by patients, including the following [58]:
 - a. patient intentionally uses/takes fewer drugs than prescribed or does not take the drug at all for whatever reason;
 - b. patient uses/takes more drugs than prescribed;
 - c. patient abuses drugs (unregulated overuse);
 - d. patient decides to use an unnecessary drug;
 - e. patient takes food that interacts;
 - f. patient stores drugs inappropriately;
 - g. inappropriate timing or dosing intervals;
 - h. patient unintentionally administers/uses the drug in the wrong way;
 - i. patient physically unable to use a drug/form as directed; and
 - j. patient is unable to understand instructions properly.

Important outcomes

1. Mortality rate
2. Worsening of NCDs
3. Clinical measurements (e.g. blood pressure and asthma control)
4. Laboratory values (e.g. LDL cholesterol, HbA1c, and PT-INR)
5. Patients' quality of life (e.g. SF-36)
6. Utilisation frequency of healthcare services (e.g. unplanned hospital admissions, and hospital and clinic visits)
7. Adverse events caused by telepharmacy
8. Economic outcomes (e.g. cost of personnel and travel, capital investment, and resource use)
9. Other healthcare providers' satisfaction

Search methods for identification of studies

Electronic searches

The Cochrane Effective Practice and Organisation of Care (EPOC) Information Specialist developed the search strategies in consultation with the review authors. Nia Wyn Roberts (Senior

Outreach Librarian, Bodleian Health Care Libraries) subsequently reviewed the search strategies. We searched the Cochrane Database of Systematic Reviews (CDSR) and Epistemonikos (www.epistemonikos.org) for related systematic reviews. We searched the following databases for primary studies:

1. Cochrane Central Register of Controlled Trials (CENTRAL; 15 December 2025), in the Cochrane Library;
2. MEDLINE Ovid (1946 to 15 December 2025);
3. Embase Ovid (1974 to 15 December 2025);
4. WHO Global Index Medicus (pesquisa.bvsalud.org/gim/advanced/?lang=en; to 15 December 2025).

Search strategies were composed of keywords and controlled vocabulary terms. We did not apply any limits on language, and we searched all databases from inception to the date of search. We used search filters where possible to limit retrieval to appropriate study designs: the Cochrane Highly Sensitive Search Strategy to identify randomised trials [59]. See [Supplementary material 1](#) for the full search strategy.

Searching other resources

Trial registries

1. WHO International Clinical Trials Registry Platform (ICTRP) (www.who.int/ictip; to 15 December 2025)
2. US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov; to 15 December 2025)

Grey literature

We conducted a grey literature search to identify studies not indexed in the databases listed above. We used the following sources.

1. OpenGrey (www.opengrey.eu; to 15 December 2025)
2. Grey Literature Report (New York Academy of Medicine; www.greylit.org; to 15 December 2025)
3. Agency for Healthcare Research and Quality (AHRQ; www.ahrq.gov; to 15 December 2025)

We conducted simplified searches using the terms: ‘telepharmacy’, ‘tele-pharmacy’, ‘telepharmacies’, and ‘tele-pharmacies’. We did not search for OpenGrey as its archive has been closed and no longer updated since summer 2021.

In addition, we assessed the reference lists of all included studies and relevant systematic reviews for additional potentially eligible primary studies. We contacted authors of included studies/reviews to clarify reported published information and to seek unpublished results/data. Researchers with expertise relevant to the review topic/EPOC interventions were contacted. We conducted cited reference searches for all studies included in the Science Citation Index, Web of Science, and screened individual journals and conference proceedings (e.g. handsearch).

In the review, we provided appendices for all strategies used, including a list of sources screened and relevant reviews/primary studies assessed [37, 49, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72].

As part of the search and screening process, we screened records for post-publication amendments, including retractions. We excluded retracted records identified during screening from the review.

Data collection and analysis

Selection of studies

We downloaded all titles and abstracts retrieved by electronic searching to a reference management database and removed duplicates [57]. Two review authors (Hideki Sugita (HS) and Miki Takenaka Sato (MTS), Takenori Ichimura (TI), Tomiko Sunaga (TS), Masayuki Maeda (MM), or Noyuri Yamaji (NY)) independently screened titles and abstracts for inclusion. We retrieved the full-text study reports/publication. Two review authors (HS and MTS, TI, TS, MM or NY) independently screened the full text to identify studies for inclusion, and recorded reasons for exclusion of the ineligible studies. We resolved any disagreement through discussion or, if required, consulted a third review author (NY or Takeshi Hasegawa (TH)).

We listed studies excluded after full-text assessment in the ‘Characteristics of excluded studies’ table. We collated multiple reports of the same study so that each study, rather than each report, was the unit of interest in the review. We also provided any information we could obtain about ongoing studies. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram [73].

Data extraction and management

We used Covidence software to facilitate the screening and data extraction process [57]. We used the Cochrane EPOC standard data collection form, and adapted it for study characteristics and outcome data [74]; we piloted the form on at least one study included in the review. Two review authors (HS and MTS, TI, TS, NY, Etsuko Nishimura (EN), Phyu Phyu Tun (PPT), or Gessica Augustin (GA)) independently extracted the following study characteristics from the included studies and entered the data into Review Manager (RevMan) [75].

1. Methods: study design, number of study centres and location, study setting, withdrawals, date of study, follow-up
2. Participants: number, mean age, age range, gender, severity of condition, diagnostic criteria, inclusion criteria, exclusion criteria, other relevant characteristics
3. Interventions: intervention components, comparison, fidelity assessment
4. Outcomes: main and other outcomes specified and collected, time points reported
5. Notes: funding for trial, notable conflicts of interest of trial authors, ethical approval

Two review authors (HS and MTS, TI, TS, NY, EN, PPT, or GA) independently extracted outcome data from included studies. We noted in the ‘Characteristics of included studies’ table if the study reported outcome data in an unusable way. We resolved disagreements by consensus or by consulting a third review author (NY, TH, or Erika Ota (EO)). One review author (HS) entered data into RevMan [75]. We double-checked the data entry for accuracy against the data extraction sheets.

Depending on the resources available, we translated studies reported in journals into languages other than English prior to

assessment. Where multiple publications existed for a single study, we combined the reports and used the publication with the most complete data in the analysis. If the relevant results were only published in an earlier version, we used those data. We highlighted where there were discrepancies between the published versions.

Risk of bias assessment in included studies

Two review authors (HS and MTS, TI, TS, NY, EN, PPT, or GA) independently assessed the risk of bias for each included study using the criteria outlined in Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* [76], and Cochrane EPOC group guidance [77]. We resolved any disagreements by discussion or by involving a third review author (NY, TH, or EO). We assessed the risk of bias according to the following domains for randomised trials:

1. bias arising from the randomisation process;
2. bias due to deviations from intended interventions;
3. bias due to missing outcome data;
4. bias in measurement of the outcome; and
5. bias in selection of the reported result.

We judged each potential source of bias as either low risk, some concerns, or high risk, and provided a quote from the study report together with a justification for our judgement in the risk of bias table. We summarised the risk of bias judgements across different studies for each of the domains listed. An overall risk of bias assessment (low risk, some concerns, or high risk) was assigned to each of the included studies using the approach suggested in Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* [76]. We considered studies with low risk of bias for all key domains, or where it seems unlikely for bias to seriously alter the results, to have a low risk of bias. We considered studies where the risk of bias in at least one domain was unclear, or judged to have some bias that could plausibly raise doubts about the conclusions, to have an unclear risk of bias. We considered studies with a high risk of bias in at least one domain, or judged to have serious bias that decreases the certainty of the conclusions, to have a high risk of bias.

For studies using cluster-randomisation, we assessed the risk of bias using the following areas [76]:

1. bias arising from the randomisation process;
2. bias arising from the timing of identification and recruitment of participants;
3. bias due to deviations from intended interventions;
4. bias due to missing outcome data;
5. bias in measurement of the outcome; and
6. bias in selection of the reported result.

We considered blinding separately for different key outcomes where necessary. Where information on risk of bias relates to unpublished data or correspondence with a trialist, we noted this in the risk of bias table. We did not exclude studies on the grounds of their risk of bias, but clearly reported the risk of bias when presenting the results of the studies. When considering treatment effects, we took into account the risk of bias in the studies that contribute to that outcome.

In line with the revised Cochrane risk-of-bias tool (RoB 2), we conducted the risk of bias assessments at the outcome level rather than solely at the study level. We assessed risk of bias for all outcomes prespecified in the review protocol, including all corresponding outcome measures and time points reported in the included studies. For each outcome, we assessed the risk of bias using the specific outcome measure applied (e.g. validated scales, clinical measurements, laboratory values, or utilisation data) at the time point closest to the end of the study or the final reported follow-up.

Measures of treatment effect

For dichotomous data (e.g. unplanned hospital admissions, and hospital and clinic visits), we estimated intervention effects using risk ratios (RRs) together with the appropriate associated 95% confidence intervals (CIs). For continuous data (e.g. blood pressure, cholesterol, and plasma glucose levels), we used mean differences (MDs) if the same measurement scales had been used, or standardised mean differences (SMDs) if different scales had been used, together with the 95% appropriate associated CI [55]. We ensured that increases in scores for continuous outcomes were interpreted consistently across studies, explained the direction to the reader, and reported where the directions were reversed, if this was necessary. We divided outcomes into dichotomous and continuous outcomes.

For medication adherence, we used SMDs. In Hoda 2023 [78], lower scores indicated better adherence, whereas in other studies, higher scores reflected better adherence. To ensure consistent direction across studies, we multiplied the mean values from Hoda 2023 by -1 before standardisation, as recommended in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [79]. We did not modify the standard deviations (SDs).

Where the outcome was only available as visual data from the figures in the article, we contacted the study authors to request the underlying data. If we did not obtain any additional data, we presented the results narratively.

For incomplete reporting of continuous outcomes measured in pretest-posttest design, we applied imputation methods described by Follmann and colleagues [80], and Abrams and colleagues [81]. For the cluster-RCTs that adopted multilevel models to address the intracluster correlations in the statistical analyses, we used the reported adjusted estimates directly in the synthesis analyses.

Unit of analysis issues

We needed to avoid unit of analysis issues occurring in cluster-RCTs when allocation occurred at the level of the cluster or the group and data were collected from individual participants. We planned to use the generic inverse variance method in RevMan to pool effect measure estimates and their standard errors where a cluster trial had been conducted and correctly analysed [75, 82]. We planned to record, as part of the risk of bias assessment, where a cluster-RCT had been conducted if incorrectly analysed. We planned to approximate the correct analyses based on the guidance in Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions*, considering the following information [82]:

1. the number of clusters (or groups) randomised for each intervention group and the total number of participants in the study; or the average (mean) size of each cluster;
2. the outcome data ignoring the cluster design for the total number of individuals; and
3. an estimate of the intracluster (or intraclass) correlation coefficient (ICC).

Where ICC estimates were not published, we contacted the study authors. For cluster-RCTs that did not adjust the intracluster correlations and ICC estimates were unavailable, we applied the effective sample size methods by Rao and Scott to adjust the design effect [83], using a conservative ICC estimate of 0.05 as recommended in Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [82].

Dealing with missing data

We contacted study authors to verify key study characteristics and to obtain missing outcome data where possible (e.g. when we identified a study as an abstract only). The relevant information obtained in this way was included in the review. We critically appraised the assumptions of the methods used to address missing data. We carefully evaluated key numerical data, such as screened patients, randomised patients, intention-to-treat, as-treated, and per-protocol populations. In addition, we assessed attrition rates, such as dropouts, losses to follow-up, and withdrawals, as recommended in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [79].

Reporting bias assessment

We attempted to contact the study authors, asking them to provide missing outcome data. Where this was not possible, and the missing data were thought to introduce serious bias, we explored the impact of including such studies in the overall assessment of the results. For the outcome of medication adherence, we were able to pool 11 trials, and therefore constructed a funnel plot to assess reporting bias. We interpreted the results with caution, as recommended in Chapter 13 of the *Cochrane Handbook for Systematic Reviews of Interventions* [84, 85].

Synthesis methods

We undertook meta-analyses and performed synthesis analyses using a random-effects model, due to possible heterogeneity in study design, population, interventions, and outcomes. If we had identified multi-arm trials meeting the eligibility criteria, we planned to pool intervention arms for a single pairwise comparison.

If we found at least two or more included studies, where we judged participants, interventions, comparisons, and outcomes to be sufficiently similar, we conducted a meta-analysis. We first assessed the heterogeneity by visual inspection of the forest plot. We used the I^2 statistic to measure heterogeneity among the trials in each analysis. We assessed I^2 values, consistent with the guidance in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [79], as follows:

1. 0% and 40%: might not be important;
2. 30% to 60%: may represent moderate heterogeneity;
3. 50% to 90%: may represent substantial heterogeneity; and

4. above 75%: represents considerable heterogeneity [79].

We considered whether there was any additional outcome information that could not be incorporated into meta-analyses, noted this in the comments, and stated if it supported or contradicted the information from the meta-analyses. If it was not possible to meta-analyse the data, we summarised the results in the text. For DRPs, we considered prevention and detection outcomes as conceptually distinct; therefore, we did not combine these outcomes in a meta-analysis and instead described the findings narratively.

Investigation of heterogeneity and subgroup analysis

We performed subgroup analyses to explore possible sources of heterogeneity using the following factors. We used a test for interaction to assess for statistically significant differences between the subgroups.

Intervention methods

We expected the intervention methods on entering the trial to have an effect on outcomes. We conducted subgroup analyses based on the four classifications of interventions [27]:

1. synchronous video (two-way interaction);
2. asynchronous content (stored audio or video);
3. remote patient monitoring; and
4. mobile health (mobile devices; telephone, e-mail, and SMS).

We evaluated primary preventive telepharmacy (e.g. smoking cessation instruction) and secondary preventive telepharmacy (e.g. antihypertensive drug management) separately. In addition, we considered interventions with a combination of multiple telepharmacies.

Geographic location

We divided geographic location into four groups according to income (i.e. low, lower-middle, upper-middle, and high income) according to the World Bank's country classification [86]. In addition, we also considered differences in places of residence (e.g. rural, urban, and inner-city), based on the PROGRESS Plus guidance [87, 88].

Equity-related assessment

We did not pre-specify equity-related characteristics in the protocol [53], and did not use a formal equity framework in this review. However, we conducted a subgroup analysis based on country income level, which may capture some equity-related aspects.

Sensitivity analysis

We evaluated the robustness of the results and its impact on effect sizes using the following sensitivity analyses.

1. Restricting the analysis to published studies
2. Restricting the analysis to studies with a low risk of bias for sequence generation and allocation concealment and incomplete outcome data
3. Restricting the analysis to cluster studies, where the allocation and the analysis were the same level
4. Missing data: we excluded trials in which missing data had been imputed

5. Restricting the analysis to studies using validated outcome measures: one study using an unvalidated measure of medication adherence was excluded in a post hoc analysis to assess the robustness of this outcome

Certainty of the evidence assessment

We created a summary of findings table for the main intervention comparison, and included the most critical/important outcomes in order to draw conclusions about the certainty of the evidence. We revised the originally planned outcomes, based on the availability and relevance of data during the review process, and explained the reasons for this in the 'Differences between protocol and review' section. The final outcomes presented in the summary of findings table were as follows.

1. Medication adherence
2. Patient satisfaction
3. DRPs, such as ADEs, and inadequate drug use by patients [58]
4. Mortality rate
5. Clinical measurements (SBP)
6. Clinical measurements (DBP)
7. Laboratory values (HbA1c)

We originally planned to include mortality rates and composite laboratory and clinical measures in the summary of findings table. Broad outcome categories, such as clinical measurements and laboratory outcomes, were disaggregated into specific indicators (e.g. blood pressure, and HbA1c) to ensure meaningful synthesis and clarity.

Two review authors (HS and NY) independently assessed the certainty of the evidence (high, moderate, low, or very low) using the five GRADE considerations (risk of bias, consistency of effect, imprecision, indirectness, and publication bias) [54]. We used methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [89], and the Cochrane EPOC worksheets [90]. In addition, we used GRADEpro software [91]. We resolved disagreements on certainty ratings through discussion. We provided justification for decisions to downgrade or upgrade the ratings using footnotes in the table, and made comments to aid readers' understanding of the review where necessary. We used plain language statements to report these findings in the review [90].

Consumer involvement

We did not involve consumers in the development or writing of this Cochrane review.

RESULTS

Description of studies

Results of the search

We performed electronic database searches in December 2025, which yielded 11,053 records. After removal of 89 duplicates, we screened 10,923 titles and abstracts and retrieved 736 full-text reports for further review. After full-text screening, we included 21 studies, identified eight ongoing studies and 19 records (corresponding to 14 studies) awaiting classification, and excluded 707 studies. A study selection flow diagram is shown in [Figure 1](#).

Figure 1. PRISMA flow diagram showing the identification, screening, eligibility assessment, and inclusion of studies for this review including ongoing studies (n = 8) and 19 records corresponding to 14 studies awaiting classification.

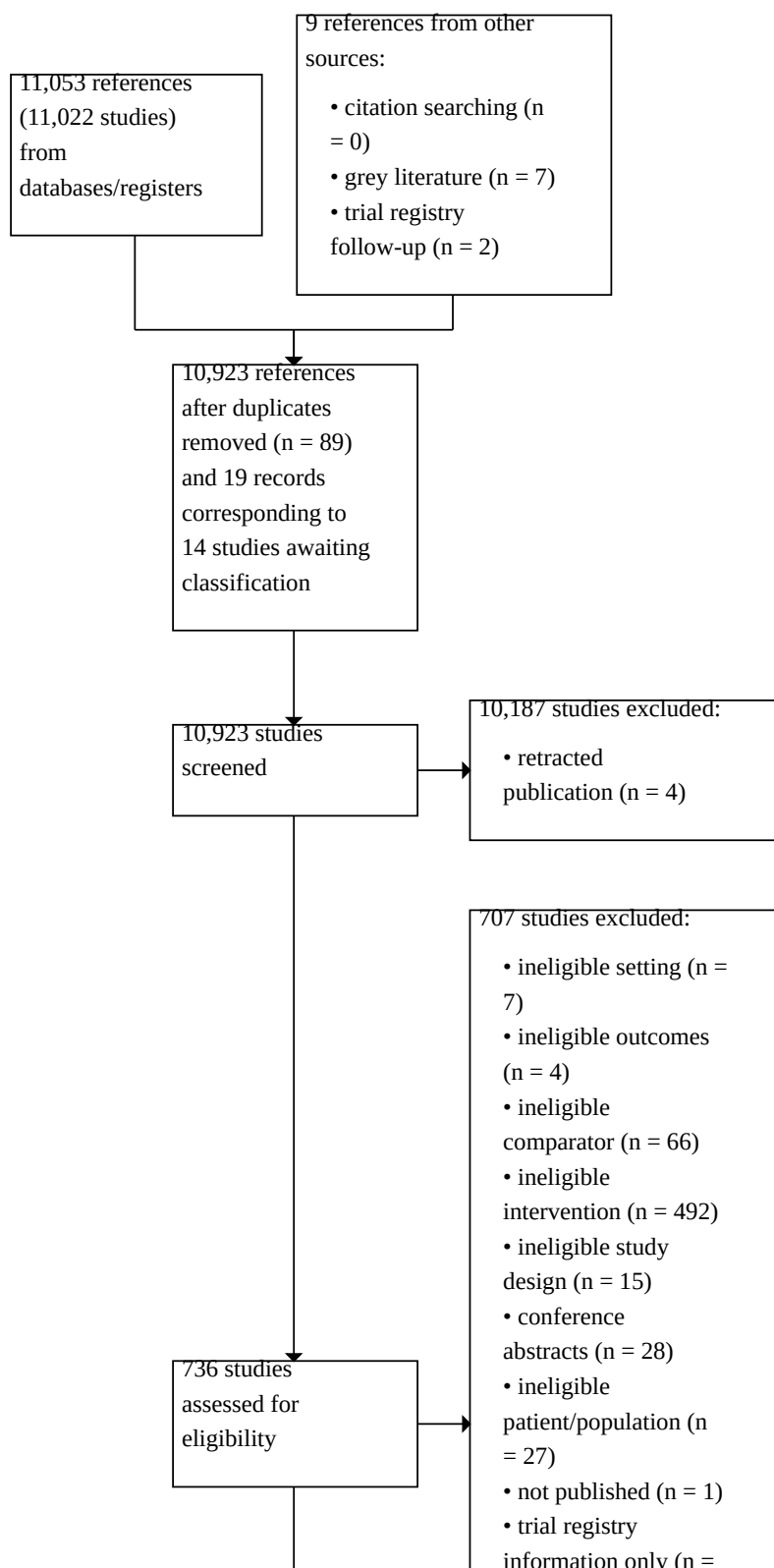
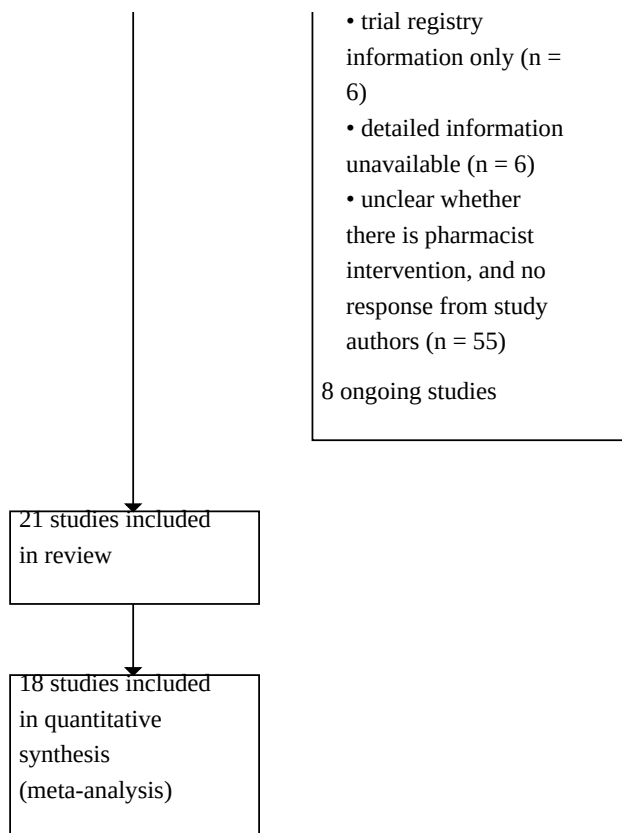


Figure 1. (Continued)



We also searched grey literature sources (Grey Literature Report and AHRQ), and screened zero and seven records, respectively. No eligible records were identified. One AHRQ record was a duplicate, which we excluded.

We attempted to contact the study investigators for 84 studies to clarify unclear or missing information relevant to eligibility. While a few study authors responded, most did not reply.

Included studies

We included 17 RCTs and four cluster-RCTs in which 5440 participants were randomised. Trial size ranged from 20 participants in Powers 1983 [92] to 1400 participants in Lauffenburger 2019 [93, 94, 95], with a median of 136 per study (interquartile range (IQR) 97 to 360). Durations of included interventions were within one month (2 to 8 days) (Bynum 2001 [96]; Powers 1983; Salmany 2018 [97]), three months (Gautier 2021 [98]; Hoda 2023; Hussain 2024 [99]; Sarayani 2018 [100, 101]; Shdaifat 2022 [102]; Sudas Na Ayutthaya 2018 [103, 104]), six months (Alsabbagh 2012 [105]; Ihekoronye 2024 [106]; Kosse 2019 [107, 108, 109, 110, 111, 112]; Mozu 2023 [113]; Young 2012 [114, 115]) and 12 months (Bosworth 2018 [116, 117, 118, 119]; Gonzales 2021 [120, 121, 122, 123, 124, 125, 126]; Ibrahim 2022 [127]; Lauffenburger 2019; Margolis 2013 [128, 129, 130, 131, 132, 133]; Odegard 2012 [134]; Scala 2018 [135, 136, 137, 138, 139]). All studies

assessed outcomes at the end of the intervention period. Margolis 2013 and Dehmer 2018 reported different outcomes from the same underlying RCT, with Dehmer 2018 presenting an economic evaluation of the intervention.

An overview of the included studies and their contribution to each outcome is presented in Table 1. Full study characteristics are available in Supplementary material 2. In addition, we provide a Supplementary summary table (Supplementary material 9), with key participant and study features (e.g. sample size, mean age) for ease of reference.

Populations

The studies were conducted in countries worldwide. Nine studies were conducted in the USA (9/21; Bosworth 2018; Bynum 2001; Gonzales 2021; Lauffenburger 2019; Margolis 2013; Odegard 2012; Powers 1983; Scala 2018; Young 2012). Two studies were conducted in Jordan (2/21; Salmany 2018; Shdaifat 2022), and one study each in Canada (Alsabbagh 2012), France (Gautier 2021), India (Hoda 2023), Pakistan (Hussain 2024), United Arab Emirates (UAE) (Ibrahim 2022), Nigeria (Ihekoronye 2024), the Netherlands (Kosse 2019), Ghana (Mozu 2023), Iran (Sarayani 2018), and Thailand (Sudas Na Ayutthaya 2018).

The studies were conducted in the following types of settings: hospital (10/21; Alsabbagh 2012; Bynum 2001; Hoda 2023; Hussain

2024; Mozu 2023; Powers 1983; Salmany 2018; Scala 2018; Shdaifat 2022; Sudas Na Ayutthaya 2018); pharmacy (6/21; Gautier 2021; Ibrahim 2022; Kosse 2019; Odegard 2012; Sarayani 2018; Young 2012); clinic (4/21; Bosworth 2018; Gonzales 2021; Ihekoronye 2024; Margolis 2013); and one study was conducted by a health insurer (1/21; Lauffenburger 2019).

The studies targeted a variety of diseases: type 2 diabetes mellitus (6/21; Gautier 2021; Hoda 2023; Ihekoronye 2024; Lauffenburger 2019; Odegard 2012; Sarayani 2018); asthma (4/21; Bynum 2001; Kosse 2019; Shdaifat 2022; Young 2012); hypertension (4/21; Ibrahim 2022; Margolis 2013; Mozu 2023; Scala 2018); cardiovascular diseases (2/21; Alsabbagh 2012; Sudas Na Ayutthaya 2018); cancer (2/21; Powers 1983; Salmany 2018); and one study each targeting hypertension or hypercholesterolaemia (Bosworth 2018), kidney failure (Gonzales 2021), and type 1 diabetes mellitus (Hussain 2024).

The mean age of participants ranged from 14.9 years in Hussain 2024 to 76.5 years in Young 2012. Seventeen out of 21 studies reported sex (81%); the proportion of female participants ranged from 15.2% to 79.7%. Seven out of nine studies conducted in the USA (78%) reported race or ethnicity; in three of these studies, participants were predominantly white (87% to 98%), and in three other studies, participants were predominantly Black or African American (50% to 91.8%).

For the purpose of subgroup analyses, we classified included studies according to the World Bank country income level at the time of study publication. Thirteen studies were conducted in HICs, one in an upper-middle-income country (UMIC) (Thailand), and seven in LMICs (India, Pakistan, Nigeria, Ghana, Iran, and Jordan). No studies were conducted in low-income countries (LICs).

Interventions and comparators

All included studies were head-to-head trials of telepharmacy versus usual care. There were various types of telepharmacy: 11 studies used telephone (Alsabbagh 2012; Bosworth 2018; Lauffenburger 2019; Mozu 2023; Odegard 2012; Powers 1983; Salmany 2018; Sarayani 2018; Scala 2018; Sudas Na Ayutthaya 2018; Young 2012); two studies used SMS (Gautier 2021; Ihekoronye 2024); and one study used video conferencing (Bynum 2001), smartphone (Gonzales 2021), telephone and SMS (Hoda 2023), mobile app (Hussain 2024), video calls and phone calls and messaging (Ibrahim 2022), smartphone app (Kosse 2019), telephone and blood pressure monitoring (Margolis 2013), and smartphone and video calling app (Shdaifat 2022). Regarding the person delivering the intervention: nine studies involved a pharmacist (including a clinical pharmacist) (Bynum 2001; Hussain 2024; Ihekoronye 2024; Mozu 2023; Odegard 2012; Powers 1983; Salmany 2018; Sarayani 2018; Shdaifat 2022); three studies involved a community pharmacist (Gautier 2021; Ibrahim 2022; Kosse 2019); two studies involved a hospital pharmacist (Scala 2018; Sudas Na Ayutthaya 2018); another two involved trained pharmacists (Hoda 2023; Young 2012); and one study involved a doctoral-level pharmacist (Margolis 2013).

The frequency of intervention varied: two studies were daily (Gautier 2021; Powers 1983); eight were weekly (Alsabbagh 2012; Gonzales 2021; Hoda 2023; Ibrahim 2022; Ihekoronye 2024; Margolis 2013; Odegard 2012; Sarayani 2018); five were monthly (Bosworth 2018; Mozu 2023; Scala 2018; Shdaifat 2022; Young 2012); two

were as needed (Hussain 2024; Kosse 2019); and one each study conducted on days 3, 7, 14, and 21 (Sudas Na Ayutthaya 2018), three sessions (Bynum 2001), between one and four calls (Lauffenburger 2019), and one call (Salmany 2018). There were various modes of delivery: 12 studies were conducted using synchronous mode alone (Alsabbagh 2012; Bosworth 2018; Bynum 2001; Lauffenburger 2019; Mozu 2023; Odegard 2012; Powers 1983; Salmany 2018; Sarayani 2018; Scala 2018; Sudas Na Ayutthaya 2018; Young 2012), two studies were conducted using mobile health combined with remote patient monitoring (Gonzales 2021; Hussain 2024), two studies were conducted using asynchronous (Gautier 2021; Ihekoronye 2024), two studies used both synchronous and asynchronous modes (Hoda 2023; Ibrahim 2022), one study used mobile health combined with synchronous delivery (Shdaifat 2022), one study used mobile health alone (Kosse 2019), and phone call synchronous and remote patient monitoring (Margolis 2013). Follow-up period was within one month (8 to 28 days) (Bynum 2001; Powers 1983; Salmany 2018), three months (Hoda 2023; Hussain 2024; Sudas Na Ayutthaya 2018), six months (Alsabbagh 2012; Gautier 2021; Ihekoronye 2024; Kosse 2019; Mozu 2023; Young 2012), nine months (Sarayani 2018), 12 months (Bosworth 2018; Gonzales 2021; Ibrahim 2022; Lauffenburger 2019; Odegard 2012; Scala 2018; Shdaifat 2022), and 18 months (Margolis 2013). Usual care typically involved face-to-face pharmacist consultations as part of routine clinical practice. Where details were not fully reported, we assumed this standard unless otherwise stated.

The interventions evaluated across studies varied widely in terms of components, including mode of delivery, technology used, frequency and intensity of contact, personnel delivering the intervention, and whether the intervention was tailored to individual participants. For clarity, we described the modes of delivery separately from the technologies used, reflecting the layered nature of telepharmacy interventions. These variations reflect differences in the design and implementation of telepharmacy interventions across settings. We have summarised the details of the intervention components for each study in [Supplementary material 10](#).

Critical outcomes

Medication adherence

Eleven trials (seven individually RCTs and four cluster-RCTs) reported medication adherence. Among these, four trials assessed adherence using the 8-item MMAS-8 (Gautier 2021; Ibrahim 2022; Ihekoronye 2024; Sarayani 2018), one trial used the 4-item MMAS (MMAS-4) (Margolis 2013), one trial used the Medication Adherence Rating Scale 5 (MARS) (Kosse 2019), one trial used the MARS 10 (Mozu 2023), two trials used the Medication Possession Ratio (MPR) (Alsabbagh 2012; Odegard 2012), and one trial used the proportion of days covered (PDC) (Lauffenburger 2019) [56, 140, 141]. One trial used MPR (Odegard 2012), but we excluded it from the meta-analysis due to unavailable total MPR data. Another study applied a non-validated adherence score (Hoda 2023), and we therefore excluded it from sensitivity analysis.

Patients' satisfaction

Five studies (four RCTs and one cluster-RCT) reported data on patients' satisfaction. Various instruments were used across studies, including the Telepharmacy Metered-Dose Inhaler Technique Evaluation instrument (Bynum 2001), a 5-point Likert scale to evaluate satisfaction with pharmaceutical care (Salmany

2018), a Visual Analogue Scale (Powers 1983), an overall health care satisfaction rating (Margolis 2013), and the Telemedicine Satisfaction Questionnaire (TSQ) (Shdaifat 2022) [142]. Of these, we excluded Salmany 2018 and Shdaifat 2022 from the meta-analysis: the former due to scale incompatibility, and the latter due to the absence of control group data.

DRPs

Six studies, all individually RCTs, reported outcomes related to DRPs. DRPs were considered a heterogeneous, primarily process-oriented outcome, encompassing both prevention of adverse events and enhanced detection of medication-related problems, depending on intervention intent. Four trials focused on ADEs, including transplant-related complications (Gonzales 2021), severe hypoglycaemic episodes (Hussain 2024), opioid-related side effects (Powers 1983), and major bleeding associated with warfarin (Sudas Na Ayutthaya 2018). One study assessed a broader range of DRPs, including inappropriate dosing and contraindications (Ibrahim 2022). One study reported medication-related adverse effects in asthma patients, but did not provide a breakdown of events by intervention and control groups (Salmany 2018). Due to heterogeneity in outcome definitions and intervention intent, we did not conduct a meta-analysis for this outcome and presented the findings narratively.

Important outcomes

Worsening of NCDs

Two RCTs reported on the outcome of worsening of NCDs. One study focused on type 1 diabetes and assessed episodes of diabetic ketoacidosis (Hussain 2024). The other addressed asthma and evaluated the frequency of acute exacerbations (Shdaifat 2022). Both trials used disease-specific outcome measures.

Clinical measurements

Ten trials (eight individually RCTs and two cluster-RCTs) reported outcomes related to clinical measurements.

Three studies assessed asthma control using validated instruments: the Control of Allergic Rhinitis and Asthma Test (CARAT) (Kosse 2019), the Childhood Asthma Control Test (C-ACT) (Shdaifat 2022), and the Asthma Control Test (ACT) (Young 2012) [143, 144, 145]. We conducted a meta-analysis using data from two of these studies; we excluded Young 2012 due to unavailability of endpoint data.

Five studies evaluated SBP and DBP as continuous outcomes (Bosworth 2018; Ibrahim 2022; Margolis 2013; Mozu 2023; Scala 2018). We conducted meta-analyses for both measurements.

Two additional studies reported heterogeneous outcomes—pain relief and thromboembolic events, respectively—which we did not include in the meta-analysis due to differences in outcome definitions and clinical relevance (Powers 1983; Sudas Na Ayutthaya 2018).

Laboratory values

Eight RCTs reported laboratory outcomes, including one cluster-RCT.

Seven studies assessed HbA1c (Bosworth 2018; Gautier 2021; Hoda 2023; Hussain 2024; Ihekoronye 2024; Lauffenburger 2019; Sarayani

2018). Of these, we excluded two studies from the meta-analysis: one reported dichotomous data (Ihekoronye 2024), and another showed inconsistencies between the table and main text that precluded accurate data extraction (Hussain 2024). We included the remaining five studies in the pooled analysis.

Two studies reported LDL cholesterol levels (Bosworth 2018; Sarayani 2018), and we included both in the meta-analysis. One study assessed INR values using a dichotomous measure of therapeutic range (Sudas Na Ayutthaya 2018). Due to the single-study reporting and outcome format, we described this result narratively.

Patients' quality of life

Three studies (two individually RCTs and one cluster-RCT) assessed patients' quality of life. Hoda 2023 evaluated quality of life in patients with type 2 diabetes using the EuroQol 5-dimensions 5-levels (EQ-5D-5L) index [146]. Kosse 2019 and Shdaifat 2022 assessed asthma-related quality of life using the Paediatric Asthma Quality of Life Questionnaire (PAQLQ) and Mini PAQLQ, respectively [147, 148].

Utilisation frequency of healthcare services

Four RCTs reported outcomes related to the utilisation of healthcare services, including hospital admissions or readmissions, and ED visits.

For hospital admissions, three studies assessed this outcome (Gonzales 2021; Salmany 2018; Sudas Na Ayutthaya 2018). Two studies provided dichotomous data (Salmany 2018; Sudas Na Ayutthaya 2018), which we included in the meta-analysis. Gonzales 2021 reported hospitalisation rates per patient and was unsuitable for pooling.

For ED visits, two studies reported outcomes (Salmany 2018; Shdaifat 2022). We excluded Salmany 2018 from meta-analysis due to insufficient extractable data, as event counts and denominators were unclear. Also, we excluded Shdaifat 2022 due to inconsistent reporting formats that were not compatible with pooling.

Economic outcomes

One RCT, Shdaifat 2022, and one cluster-RCT, Dehmer 2018 (as part of Margolis 2013), reported economic outcomes. Considering the types of cost-related outcomes analysed were varied, we deemed these studies unsuitable for pooled analysis.

No studies reported outcomes on the mortality rate, adverse events caused by telepharmacy, or the satisfaction of other healthcare providers. As no adverse events attributable to telepharmacy were reported, potential harms remain uncertain.

Ongoing studies

We identified eight ongoing studies for which no results have been published. The conditions targeted in these studies included asthma, cancer, chronic obstructive pulmonary disease (COPD), hypertension, kidney transplant recipients, rheumatic diseases, type 2 diabetes mellitus, anticoagulant therapy, and studies targeting multiple chronic conditions. We provided the details of these ongoing studies in the 'Characteristics of ongoing studies' table (Supplementary material 5).

Studies awaiting classification

Nineteen records (corresponding to 14 studies) had available results (e.g. conference abstracts and full-text publications). However, we could not fully assess these at the time of this review, and we therefore classified them as awaiting classification. We provided the reasons for this classification in the 'Characteristics of studies awaiting classification' tables ([Supplementary material 4](#)).

Excluded studies

We excluded 707 studies during full-text screening. The primary reason for exclusion was ineligible intervention, accounting for 492 studies. Additionally, we excluded 55 studies due to uncertainty regarding whether the intervention was delivered by pharmacists, as we did not receive a response from the study authors. A representative list of excluded studies and the reasons for their exclusion are in [Supplementary material 3](#).

Risk of bias in included studies

We have included risk of bias assessments and support for judgements across all RoB 2 domains for each outcome that we prespecified for risk of bias assessments in the review protocol [53] (see [Supplementary material 6](#)). For the four cluster-RCTs (Gautier 2021; Kosse 2019; Margolis 2013; Odegard 2012), we detailed the cluster-RCT-specific risk of bias assessments and support for judgements in the overall risk of bias column. Full consensus responses to the signalling questions for each domain, across all studies and outcomes, are available from the Open Science Framework (OSF; DOI: 10.17605/OSF.IO/KBNZ3).

Across the 21 included studies, there was considerable variation in the risk of bias judgments across outcomes and outcome measures. For different outcomes, studies were distributed among low risk, some concerns, and high risk of bias, indicating inconsistencies in methodological quality throughout the evidence base.

Critical outcomes

Medication adherence

Eleven trials reported medication adherence outcomes (Alsabbagh 2012; Gautier 2021; Hoda 2023; Ibrahim 2022; Ihekoronye 2024; Kosse 2019; Lauffenburger 2019; Margolis 2013; Mozu 2023; Odegard 2012; Sarayani 2018). Across included studies, we judged the risk of bias arising from the randomisation process as low in six studies, with some concerns in four studies, and one cluster-RCT judged to be at high risk of bias (Odegard 2012). We generally assessed individually RCTs with clearly described randomisation and allocation concealment as low risk, whereas minor baseline imbalances or limited reporting contributed to some concerns, particularly in cluster-RCTs.

All studies were unblinded due to the nature of the telepharmacy interventions. We judged the risk of bias due to deviations from intended interventions as low or some concerns in most studies. We judged two studies to be at high risk of bias because analyses deviated from the effect of assignment to intervention.

Bias due to missing outcome data was judged as low risk in most studies, with attrition generally below 5% and similar between groups. However, we considered three studies to be at high risk of bias due to substantial loss to follow-up, imbalances in missing

data between groups, or insufficient information on the handling of missing outcome data.

For bias in measurement of the outcome, we judged eight studies to be at high risk of bias, as medication adherence was predominantly assessed using self-reported measures. In contrast, we considered three studies using objective adherence measures based on pharmacy dispensing data to be at low risk of bias.

For bias in selection of the reported result, we judged four studies to be at low risk of bias because pre-specified protocols or registry information were available. In contrast, we assessed seven studies to be at high risk of bias due to the absence of a pre-specified analysis plan or post hoc selection of adherence outcomes or analyses.

Overall, the risk of bias for medication adherence outcomes was most frequently judged to be at high risk of bias. This was primarily driven by high risk of bias in outcome measurement and selection of the reported result, largely due to reliance on self-reported adherence measures and limited pre-specification of analytical approaches.

Patients' satisfaction

Four studies reported patients' satisfaction outcomes (Bynum 2001; Margolis 2013; Powers 1983; Salmany 2018). Across included studies, we generally judged the risk of bias arising from the randomisation process as either low or some concerns. In contrast, we frequently assessed the bias due to missing outcome data and selection of the reported result to be at high risk of bias. Across studies, blinding was not feasible due to the nature of the telepharmacy interventions. We judged the risk of bias due to deviations from intended interventions as high in two studies and as some concerns in two studies, reflecting per-protocol analyses that deviated from the effect of assignment to an intervention. For bias in measurement of the outcome, we considered three studies to be at high risk of bias because unblinded studies assessed patients' satisfaction using self-reported measures.

Overall, we judged the risk of bias for patients' satisfaction outcomes to be at high risk of bias across studies, primarily driven using self-reported satisfaction measures.

DRPs

Five individually RCTs reported outcomes related to DRPs (Gonzales 2021; Hussain 2024; Ibrahim 2022; Powers 1983; Sudas Na Ayutthaya 2018). Across included studies, in general, we judged the risk of bias arising from the randomisation process as low. In contrast, we frequently assessed the bias due to missing outcome data and selection of the reported result to be at high risk of bias. Across studies, blinding was not feasible due to the nature of the telepharmacy interventions. We judged the risk of bias due to deviations from intended interventions as low in three studies, with one study judged to have some concerns and one study judged to be at high risk of bias. For bias in measurement of the outcome, we assessed three studies as at high risk of bias when trials assessed DRPs or ADEs using patient self-reports or subjective judgement. In contrast, we judged two studies using structured definitions or objective clinical records to be at low risk of bias.

Overall, we considered the risk of bias for DRPs outcomes to be at high risk of bias in most studies, primarily driven by high risk of bias in outcome measurement.

Important outcomes

Worsening of NCDs

Two trials reported outcomes related to worsening of NCDs (Shdaifat 2022; Hussain 2024). Across included trials, we judged the risk of bias arising from the randomisation process and missing outcome data as low or some concerns. In contrast, we assessed bias due to selection of the reported result to be high in one study. Across studies, blinding was not feasible due to the nature of the telepharmacy interventions. We judged the risk of bias due to deviations from intended interventions as low in one study and as some concerns in the other study, reflecting differences in analytical approaches and reporting of deviations. For bias in measurement of the outcome, we assessed one trial to be at high risk of bias because the trial assessed worsening of NCDs using measures that included self-reported components, whereas we judged another study to be at low risk of bias as the trial defined outcomes using objective clinical events based on medical records.

Overall, the risk of bias for worsening of NCDs outcomes varied between the two studies, as we judged one study as at high risk of bias and the other as at low risk of bias.

Clinical measurements

Ten trials reported outcomes related to clinical measurements, comprising eight individually RCTs and two cluster-RCTs (Bosworth 2018; Ibrahim 2022; Kosse 2019; Margolis 2013; Mozu 2023; Powers 1983; Scala 2018; Shdaifat 2022; Sudas Na Ayutthaya 2018; Young 2012). These studies assessed a range of clinical measurements, including asthma control, SBP, DBP, pain relief, and thromboembolic events.

Across included studies, in general, we judged the risk of bias arising from the randomisation process as either low or some concerns. Across studies, we generally assessed the risk of bias due to deviations from intended interventions as either low or some concerns. We considered two studies to be at high risk of bias due to limited reporting on deviations and use of per-protocol or available-case analyses. Bias due to missing outcome data was mostly judged as low or some concerns, although we judged one study to be at high risk of bias. For bias in measurement of the outcome, we assessed five studies assessing blood pressure using standardised and objective measurement procedures to be at low risk of bias, whereas we assessed five studies relying on self-reported measures, including asthma control questionnaires or patient-reported pain and events, to be at high risk of bias. Bias due to selection of the reported result varied across studies, with judgements ranging from low risk to high risk of bias, reflecting differences in the availability of pre-specified analysis plans and outcome reporting.

Overall, we judged the risk of bias for clinical measurement outcomes to be either high or some concerns across studies. This judgement was primarily driven by a subjective outcome assessment for asthma control, pain, and patient-reported events, whereas we more commonly judged studies assessing blood pressure using objective measurement procedures to have some concerns rather than high risk of bias.

Laboratory values

Eight RCTs reported outcomes related to laboratory values, including one cluster-RCT (Bosworth 2018; Gautier 2021; Hoda 2023; Hussain 2024; Ihekoronye 2024; Lauffenburger 2019; Sarayani 2018; Sudas Na Ayutthaya 2018). These studies assessed laboratory outcomes including HbA1c, LDL cholesterol, and PT-INR. In general, across the included studies, we judged the risk of bias arising from the randomisation process as either low or some concerns.

Across studies, the risk of bias due to deviations from intended interventions was generally judged as low or some concerns. Most studies conducted analyses broadly consistent with the effect of assignment to intervention, with low or balanced attrition. However, we judged three studies to be at high risk of bias due to the use of per-protocol analyses in the presence of post-randomisation exclusions or incomplete reporting of deviations. We assessed bias due to missing outcome data as either low or some concerns in most studies, with one study judged to be at high risk of bias due to incomplete outcome data. For bias in measurement of the outcome, we judged all studies to be at low risk of bias. The trials measured HbA1c, LDL cholesterol, and PT-INR using standardised and objective laboratory procedures applied consistently across study arms, and we considered these unlikely to be influenced by lack of blinding. We also judged bias due to selection of the reported result as low or some concerns in most studies, with one study judged to be at high risk of bias due to the absence of a pre-specified analysis plan.

Overall, the risk of bias for HbA1c outcomes varied across studies, ranging from low to high, with higher risk primarily driven by deviations from intended interventions and missing outcome data. For LDL cholesterol, we judged the overall risk of bias as some concerns or high, reflecting similar issues related to deviations from intended interventions. We assessed the one study reporting PT-INR as having some concerns overall, mainly due to selective reporting and missing outcome data.

Patients' quality of life

Three trials reported outcomes related to patients' quality of life, comprising two individually RCTs and one cluster-RCT (Hoda 2023; Kosse 2019; Shdaifat 2022). Across included studies, we judged the risk of bias arising from the randomisation process to be low or some concerns. Across studies, the risk of bias due to deviations from intended interventions was judged as some concerns in all studies, primarily due to the lack of blinding and reliance on per-protocol analyses, with limited reporting on the handling of deviations. We assessed the bias due to missing outcome data as low or some concerns, mainly due to post-randomisation exclusions and limited reporting of missing data handling. For bias in measurement of the outcome, we judged all studies to be at high risk of bias because the unblinded studies assessed quality of life using self-reported questionnaires. Bias due to selection of the reported result varied across trials, as we judged one study as at low risk of bias and the remaining two studies at high risk due to the absence of pre-specified protocols or analysis plans.

Overall, we considered the risk of bias for patients' quality of life outcomes to be high across studies, primarily driven by subjective, self-reported outcome assessment in unblinded studies.

Utilisation frequency of healthcare services

Four RCTs reported the outcomes related to utilisation of healthcare services (Gonzales 2021; Salmany 2018; Shdaifat 2022; Sudas Na Ayutthaya 2018). These studies assessed hospital admissions or readmissions, and ED visits.

Across included studies, we judged the risk of bias arising from the randomisation process as low in most studies, with some concerns in one study. Across studies, we assessed the risk of bias due to deviations from intended interventions as low or some concerns. For ED visits, judgements were more commonly rated as some concerns, reflecting limited reporting on deviations and lack of blinding. We judged bias due to missing outcome data as low or some concerns across trials. For bias in measurement of the outcome, in general, we judged the risk of bias as low. Most studies used objective definitions and healthcare records to ascertain hospital admissions and ED visits, whereas we judged one study to be at high risk of bias because outcome assessment partly relied on patient self-report. Bias due to selection of the reported result varied across studies, with judgements ranging from low to high risk of bias.

Overall, the risk of bias for utilisation of healthcare services outcomes varied across studies, ranging from low to high. For hospital admissions, the higher risk of bias was mainly driven by concerns related to outcome measurement, missing outcome data,

and selection of the reported result. More commonly, for ED visits, we considered the overall risk of bias as some concerns or high, reflecting issues related to deviations from intended interventions, incomplete outcome data, and selective reporting.

Economic outcomes

Two RCTs reported economic outcomes (Dehmer 2018 (as part of Margolis 2013); Shdaifat 2022). We did not formally assess the risk of bias for this outcome using the RoB 2 tool because of substantial heterogeneity in outcome definitions and reporting across these studies. Therefore, the findings should be interpreted with caution.

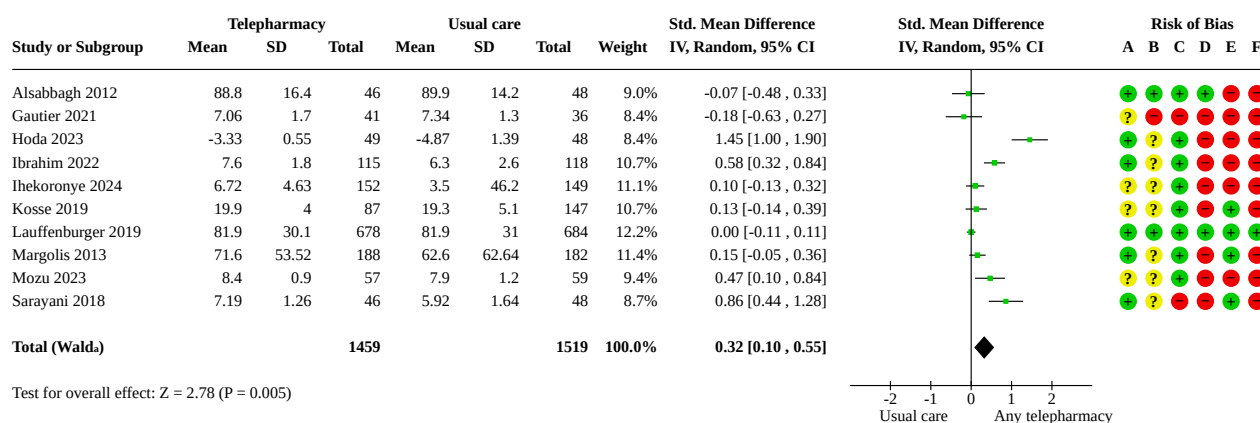
Synthesis of results

Critical outcomes

1.1 Medication adherence

A meta-analysis of 10 studies (2978 participants) showed an improvement in medication adherence among participants in the telepharmacy group compared with usual care (SMD 0.32, 95% CI 0.10 to 0.55; $P = 0.005$, $I^2 = 87\%$; 10 studies, 2978 participants; low-certainty evidence; Figure 2). One additional study reported results narratively and was not included in the meta-analysis. Subgroup analysis by intervention category, telepharmacy approach, and country income level showed no differences between those groups, with P values of 0.10, 0.16, and 0.06, respectively (Analysis 2.1, Analysis 2.2, and Analysis 2.3 in Supplementary material 7).

Figure 2. Forest plot showing the effect of telepharmacy compared with usual care on medication adherence.



Footnotes

^aCI calculated by Wald-type method.

^b Tau^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

We excluded one additional study, Odegard 2012, from the meta-analysis due to insufficient reporting of group-level MPR data, which reported improved medication adherence in the intervention group. Specifically, MPR increased from 0.85 to 0.90 over 12 months, while remaining unchanged or slightly declining

in the control group. A difference-in-difference analysis indicated a between-group benefit at 12 months, but not at six months.

1.1.1 Medication adherence: sensitivity analysis – overall risk of bias

As part of a sensitivity analysis, we re-ran the analysis after excluding studies with some concerns and a high risk of bias. We only judged one study, Lauffenburger 2019, to be at low risk of bias across all domains (MD 0.00, 95% CI -3.25 to 3.25; $P = 1.00$; 1 study, 1362 participants; Analysis 2.4 in [Supplementary material 7](#)).

1.1.2 Medication adherence: sensitivity analysis – unit of analysis matches unit of allocation

Excluding the cluster-RCT, Gautier 2021, that did not account for clustering in the analysis increased the effect size (SMD 0.37, 95% CI 0.13 to 0.61; $P = 0.003$, $I^2 = 88\%$; 9 studies, 2901 participants; Analysis 2.5 in [Supplementary material 7](#)), consistent with the main analysis.

1.1.3 Medication adherence: sensitivity analysis – imputation of missing outcome data

When we excluded studies that had imputed outcome data (Lauffenburger 2019), the effect on medication adherence

remained similar (SMD 0.37, 95% CI 0.11 to 0.63; $P = 0.006$, $I^2 = 84\%$; 9 studies, 1616 participants; Analysis 2.6 in [Supplementary material 7](#)).

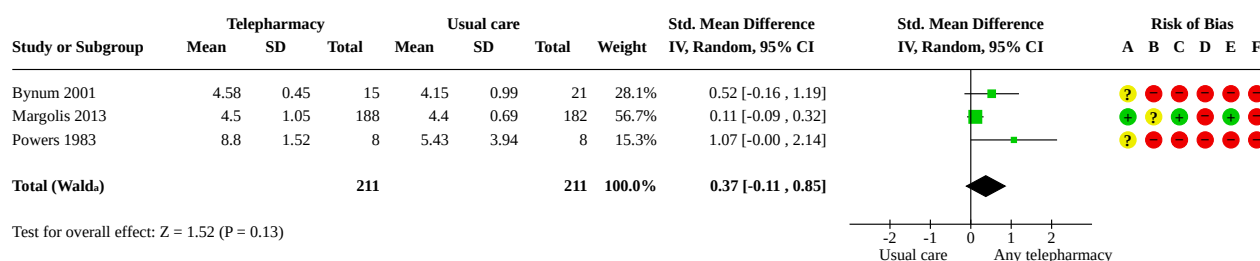
1.1.4 Medication adherence: sensitivity analysis – validity of outcome measures

When we excluded studies that used unvalidated outcome measures (Hoda 2023), the effect on medication adherence was attenuated. However, a difference remained (SMD 0.21, 95% CI 0.04 to 0.39; $P = 0.02$, $I^2 = 77\%$; 9 studies, 2881 participants; Analysis 2.7 in [Supplementary material 7](#)).

1.2 Patients' satisfaction

A meta-analysis of three studies (422 participants) showed no clear difference in patients' satisfaction between telepharmacy and usual care (SMD 0.37, 95% CI -0.11 to 0.85; $P = 0.13$, $I^2 = 50\%$; 3 studies, 422 participants; low-certainty evidence; [Figure 3](#)).

Figure 3. Forest plot showing the effect of telepharmacy compared with usual care on patients' satisfaction with care.



Footnotes

^aCI calculated by Wald-type method.

^b τ^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

We excluded two studies from the meta-analysis due to differences in outcome measurement. One study used a category-based 5-point Likert scale (from “not satisfied at all” to “very satisfied”) (Salmany 2018), and reported no clear between-group difference in satisfaction (96.5% in the intervention group versus 97.5% in the control group). The other study assessed satisfaction only in the intervention group (Shdaifat 2022), we therefore excluded it from synthesis.

1.3 DRPs

Six studies reported outcomes related to DRPs. However, we did not conduct a meta-analysis due to substantial heterogeneity in outcome definitions, measurement methods, and intervention goals.

Ibrahim 2022 aimed to improve the detection of DRPs using a broader definition that included inappropriate dosing, contraindications, and allergy risks. The authors reported a higher

number of patients with at least one DRPs in the intervention group (41/115, 35.7%) than in the control group (27/118, 22.9%).

In contrast, four trials assessed ADEs with the primary aim of reducing their incidence (Gonzales 2021; Hussain 2024; Powers 1983; Sudas Na Ayutthaya 2018), reflecting prevention of DRPs. Gonzales 2021 reported a slightly lower adverse event score in the intervention group (mean 20.68, SD 18.35) than in the control group (mean 21.26, SD 25.24), with no clear difference between groups ($P = 0.83$). Hussain 2024 found fewer severe hypoglycaemic episodes in the intervention group (mean 5.0, SD 1.4) compared to the control group (mean 9.0, SD 2.0). Powers 1983 found a comparable number of opioid-related side effects between the intervention (mean 1.63, SD 1.60; 8 participants) and control groups (mean 1.5, SD 1.41; 8 participants), though the study authors noted descriptively fewer side effects in the intervention group. No statistical comparison was reported. Sudas Na Ayutthaya 2018 reported more major bleeding events in the intervention group

(5/25, 20%) than in the control group (1/25, 4%), but the difference was not considered clinically meaningful.

Due to differences in outcome definitions (ADEs versus broader DRPs) and intervention intent (prevention versus detection), we did not consider the studies sufficiently comparable for meta-analysis. Therefore, we presented the findings narratively. An additional study reported adverse effects in 20% of participants (Salmay 2018), but did not provide group-level data, precluding comparative analysis. The certainty of the evidence for this outcome was low.

Important outcomes

1.4 Worsening of NCDs

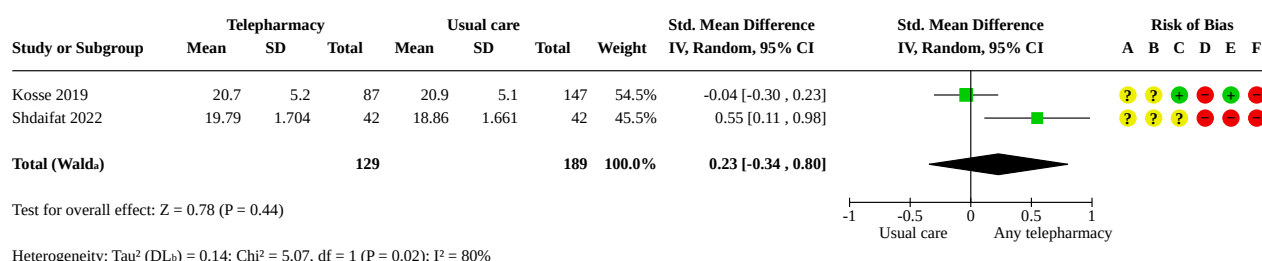
Two studies (196 participants) reported on clinically important worsening of NCDs following telepharmacy, including episodes

of diabetic ketoacidosis in people with type 1 diabetes and acute asthma exacerbations. Given the clear clinical heterogeneity between conditions and outcome definitions, we did not pool these findings and reported them narratively. The limited number of studies precludes firm conclusions regarding the effect of telepharmacy on preventing disease worsening.

1.5 Clinical measurements (asthma control)

A meta-analysis of two studies showed no clear difference in asthma control between telepharmacy and usual care (SMD 0.23, 95% CI -0.34 to 0.80; $P = 0.44$, $I^2 = 80\%$; 2 studies, 318 participants; [Figure 4](#)).

Figure 4. Forest plot showing the effect of telepharmacy compared with usual care on asthma control.



Footnotes

^aCI calculated by Wald-type method.

^b Tau^2 calculated by DerSimonian and Laird method.

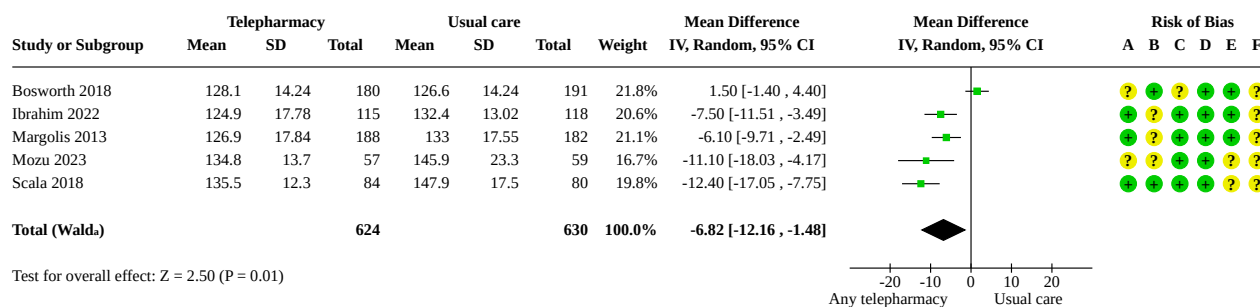
Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

One study reported ACT scores only in graphical form (Young 2012), and we could not extract or obtain numerical data from the study authors; therefore, we excluded it from the meta-analysis. The study authors reported similar ACT scores between groups at three months post-intervention. However, within the intervention group, ACT scores improved from baseline to the third telephone follow-up.

1.6 Clinical measurements (SBP)

A meta-analysis of five studies showed that telepharmacy was associated with a reduction in SBP compared to usual care (MD -6.82 mmHg, 95% CI -12.16 to -1.48; $P = 0.01$, $I^2 = 88\%$; 5 studies, 1254 participants; low-certainty evidence; [Figure 5](#)). Subgroup analysis by telepharmacy approach and country income level showed no differences between those groups, with P values of 0.94 and 0.27, respectively (Analysis 2.8 and Analysis 2.9 in [Supplementary material 7](#)).

Figure 5. Forest plot showing the effect of telepharmacy compared with usual care on systolic blood pressure.**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

(A) Bias arising from the randomization process

(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

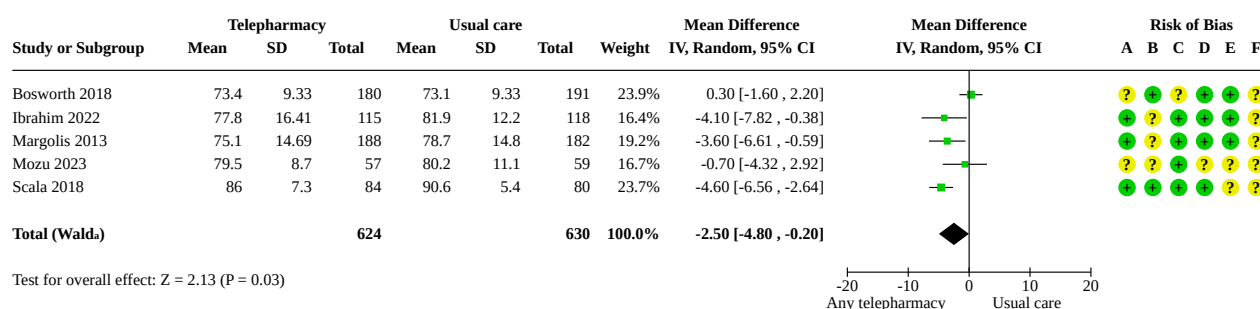
(F) Overall bias

1.6.1 Clinical measurements (SBP): sensitivity analysis – imputation of missing outcome data

Only one study remained after excluding studies with imputed outcome data (Bosworth 2018), showing no clear effect on SBP (MD 1.50 mmHg, 95% CI -1.40 to 4.40; $P = 0.31$; 1 study, 371 participants; Analysis 2.10 in [Supplementary material 7](#)).

1.7 Clinical measurements (DBP)

A meta-analysis of five studies showed telepharmacy was associated with a reduction in DBP compared with usual care (MD -2.50 mmHg, 95% CI -4.80 to -0.20; $P = 0.03$, $I^2 = 73\%$; 5 studies, 1254 participants; low-certainty evidence; [Figure 6](#)). Subgroup analysis by telepharmacy approach and country income level showed no differences between those groups ($P = 0.33$ and $P = 0.34$, respectively; Analysis 2.11 and Analysis 2.12 in [Supplementary material 7](#)).

Figure 6. Forest plot showing the effect of telepharmacy compared with usual care on diastolic blood pressure.**Footnotes**^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

(A) Bias arising from the randomization process

(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

(F) Overall bias

1.7.1 Clinical measurements (DBP): sensitivity analysis – imputation of missing outcome data

Only one study remained after excluding studies with imputed outcome data (Bosworth 2018), showing no clear effect on DBP (MD 0.30 mmHg, 95% CI –1.60 to 2.20; $P = 0.76$; 1 study, 371 participants; Analysis 2.13 in [Supplementary material 7](#)).

1.8 Clinical measurements (pain control and thromboembolic events)

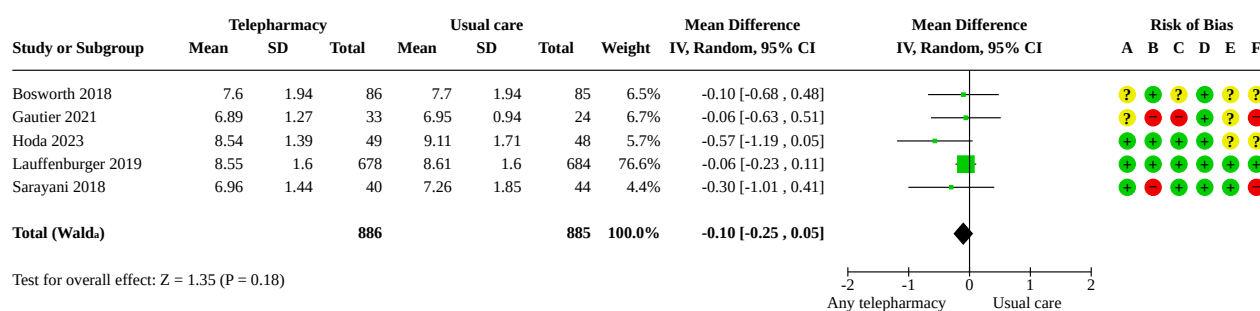
Two studies assessed pain control and thromboembolic events, but we did not include them in a meta-analysis due to limited data. Powers 1983 reported higher pain relief in the intervention group (mean VAS score 9.0, SD 1.41) than in the control group (mean

6.61, SD 3.83). Sudas Na Ayutthaya 2018 reported thromboembolic events in 12% of the intervention group versus 24% in the control group.

1.9 Laboratory values (HbA1c)

A meta-analysis of five studies showed no clear difference in HbA1c levels between telepharmacy and usual care (MD –0.10%, 95% CI –0.25 to 0.05; $P = 0.20$, $I^2 = 0\%$; 5 studies, 1771 participants; moderate-certainty evidence; [Figure 7](#)). Subgroup analyses by intervention method category, telepharmacy approach, and country income level showed no clear differences between groups ($P = 0.88$, 0.13, and 0.12, respectively; Analysis 2.14, Analysis 2.15, and Analysis 2.16 in [Supplementary material 7](#)).

Figure 7. Forest plot showing the effect of telepharmacy compared with usual care on HbA1c levels.



Footnotes

aCI calculated by Wald-type method.

bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

We excluded one study from the analysis due to unresolved discrepancies between the HbA1c values reported in the main text and those in the data table (Hussain 2024). We were unable to obtain any clarification from the trial authors.

In addition, we excluded one additional study (Ihekoronye 2024), which reported HbA1c as a categorical outcome, from the meta-analysis. This study found a higher proportion of patients achieving glycaemic control in the telepharmacy group than in the usual care group.

1.9.1 Laboratory values (HbA1c): sensitivity analysis – overall risk of bias

We only judged one study, Lauffenburger 2019, to be at low risk of bias across all domains (MD –0.06, 95% CI –0.23 to 0.11; $P = 0.49$; 1 study, 1362 participants; Analysis 2.17 in [Supplementary material 7](#)).

1.9.2 Laboratory values (HbA1c): sensitivity analysis – unit of analysis matches unit of allocation

As cluster adjustment was not performed in Gautier 2021, we excluded this study from this analysis. Excluding this cluster-RCT

did not materially change the effect estimate (MD –0.11, 95% CI –0.26 to 0.05; $P = 0.17$, $I^2 = 0\%$; 4 studies, 1677 participants; Analysis 2.18 in [Supplementary material 7](#)), consistent with the main analysis.

1.9.3 Laboratory values (HbA1c): sensitivity analysis – imputation of missing outcome data

When we excluded studies with imputed outcome data, the effect on HbA1c levels remained similar (MD –0.06, 95% CI –0.23 to 0.10; $P = 0.45$, $I^2 = 0\%$; 2 studies, 1533 participants; Analysis 2.19 in [Supplementary material 7](#)), consistent with the main analysis.

1.10 Laboratory values (LDL cholesterol)

A meta-analysis of two studies showed no clear difference in LDL cholesterol values between telepharmacy and usual care (MD –0.84 mg/dL, 95% CI –4.70 to 3.02; $P = 0.67$, $I^2 = 0\%$; 2 studies, 444 participants; Analysis 1.7 in [Supplementary material 7](#)).

1.11 Laboratory values (PT-INR)

As only one trial assessed this outcome (Sudas Na Ayutthaya 2018), we presented the results narratively. Sudas Na Ayutthaya 2018 found a higher proportion of PT-INR values within the therapeutic

range in the intervention group (45.6%) than in the control group (24.1%; $P = 0.005$).

1.12 Patients' quality of life

Three studies assessed patients' quality of life using different measurement tools: the EQ-5D-5L index (Hoda 2023), PAQLQ (Kosse 2019), and the Mini PAQLQ (Shdaifat 2022). We initially conducted a meta-analysis, which showed no clear effect of telepharmacy compared to usual care (SMD 0.38, 95% CI -0.45 to 1.21; $P = 0.37$, $I^2 = 93\%$; 3 studies, 415 participants; Analysis 1.8 in [Supplementary material 7](#)). However, due to substantial clinical and methodological heterogeneity in measurement instruments and populations, we judged that the studies were not sufficiently comparable and presented the findings narratively.

Hoda 2023 reported an improvement in the EQ-5D-5L index within the intervention group ($P = 0.024$), but no between-group comparison was available. Kosse 2019 found no important difference in PAQLQ total scores between groups (effect size 0.03, 95% CI -0.13 to 0.20). Shdaifat 2022 reported a greater improvement in Mini PAQLQ scores in the intervention group compared to the control group at the end of the study.

1.13 Utilisation frequency of healthcare services (hospital admission)

We included two studies in a meta-analysis (Salmany 2018; Sudas Na Ayutthaya 2018), which showed no clear difference between telepharmacy and usual care (RR 0.84, 95% CI 0.64 to 1.11; $P = 0.22$, $I^2 = 0\%$; 2 studies, 382 participants; Analysis 1.9 in [Supplementary material 7](#)). In Salmany 2018, the number analysed per group was unclear. We used the number randomised (166 per group), as the study authors did not respond to queries.

One study reported total hospitalisations per group (intervention: 44; control: 74) and a lower hospitalisation rate in the intervention group (Gonzales 2021). However, we excluded this study from the meta-analysis due to an incompatible outcome format.

1.14 Utilisation frequency of healthcare services (ED visits)

Two studies reported ED visit rates. Shdaifat 2022 showed a reduction in telepharmacy (38.1% versus 61.9%). Salmany 2018 found no clear difference (44% versus 52%), but event counts and denominators were unclear and unverifiable. Due to differences in reporting formats and a lack of extractable data, we could not conduct a meta-analysis.

1.15 Economic outcomes

Meta-analysis was not feasible due to heterogeneity in economic outcome reporting. We therefore narratively summarised the results from two RCTs.

The USA-based study (Dehmer 2018 (as part of Margolis 2013)) compared home blood pressure telemonitoring with pharmacist case management versus usual care in patients with uncontrolled

hypertension [130]. Over 12 months, the mean total medical cost was USD 281.40 lower in the intervention group (95% CI -3133 to 2550). Hypertension-related pharmacy costs increased (MD USD 126.00, 95% CI 64 to 188), while lipid-related pharmacy costs decreased (MD USD 44.20, 95% CI -87 to -1). Clinic, laboratory, and hospital costs were similar between groups.

A trial in Jordan assessed pharmacist-led telemedicine for paediatric asthma over 12 weeks (Shdaifat 2022). Costs were reported in Jordanian Dinars (1 JD = 1.4 USD). Compared with usual care, the intervention group had lower private medication costs (13.16 JD versus 26.74 JD), transportation costs (median 11 JD versus 17 JD), and clinic visit costs (median 5 JD versus 16 JD). Hospital burden decreased more in the intervention group (from 85.64 JD to 40.88 JD) than in the control group (from 83.70 JD to 77.86 JD). At the study end, parental wage loss was lower in the intervention group (39.51 JD versus 55.44 JD).

No studies provided data on mortality, telepharmacy-related adverse events, or the satisfaction of other healthcare providers. We did not analyse these outcomes.

We were unable to conduct subgroup and sensitivity analyses for some outcomes – patients' satisfaction, DRPs, and clinical measurements (asthma control) – due to the limited number of included studies (< three trials), lack of relevant subgroups, or overlap with the main analysis. In contrast, we performed these analyses (as prespecified in the protocol for other outcomes), such as medication adherence, SBP and DBP, and HbA1c. However, we did not observe any clear subgroup differences, and sensitivity analyses yielded results consistent with the main findings. All studies were published, so this sensitivity analysis was not applicable. We were unable to conduct subgroup analyses based on place of residence (e.g. rural, urban, inner-city), which we pre-specified in the protocol as part of equity considerations, due to lack of relevant data in the included studies. Details are available in [Supplementary material 7](#).

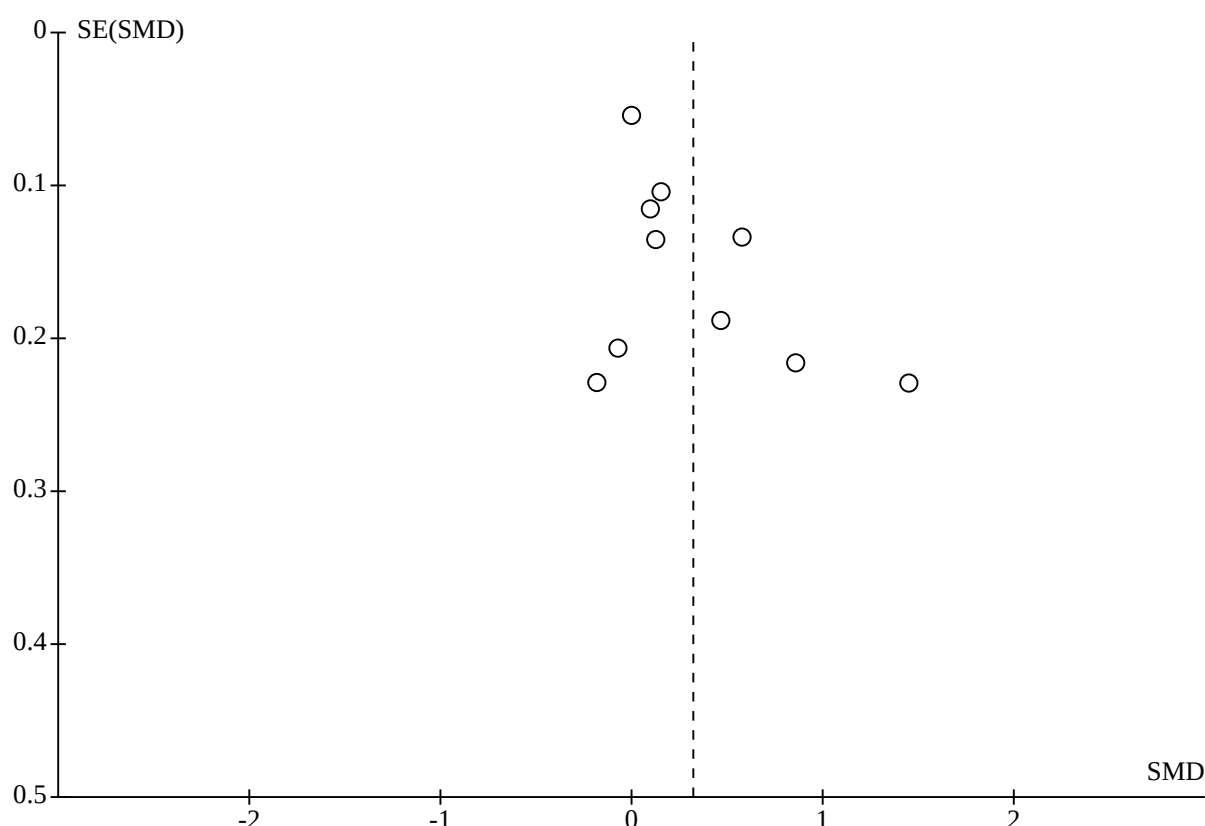
Equity assessment

We assessed equity-related factors using the PROGRESS-Plus framework [87]. While some studies reported race or ethnicity as baseline characteristics, none evaluated treatment effect differences by these factors. We conducted subgroup analyses by country income level for four outcomes (medication adherence, SBP, DBP, HbA1c), but none showed clear subgroup differences.

Reporting biases

We generated a funnel plot for the critical outcome medication adherence, which included 10 studies ([Figure 8](#)). The distribution of effect estimates appeared symmetrical around the pooled effect, with no clear indication of asymmetry. This suggests that small-study effects or publication bias were unlikely to have substantially influenced the result. We did not produce funnel plots for other outcomes or the sensitivity analyses, due to the limited number of studies contributing to each analysis.

Figure 8. Funnel plot of studies reporting medication adherence (n = 10). The vertical axis shows the standard error of the effect estimate; the horizontal axis shows the mean difference. The distribution appeared symmetrical, suggesting no strong indication of publication bias.



DISCUSSION

Summary of main results

This review included 21 randomised trials (17 individually RCTs and four cluster-RCTs), involving 5440 participants with NCDs in ambulatory care settings. The trials evaluated a wide range of pharmacist-led telepharmacy interventions, varying in frequency, intensity, content, and delivery mode. Interventions were delivered via telephone calls, video consultations, mobile apps, web platforms, and asynchronous messaging systems. Comparator groups typically received the usual care, including standard medical management and general pharmacist support where available. Follow-up periods ranged from eight days to 18 months.

We grouped outcomes into the following nine domains, with subcategories reflecting how we analysed outcomes in this review: medication adherence; patients' satisfaction; DRPs; worsening of NCDs; clinical measurements (asthma control, SBP, DBP, pain control, thromboembolic events); laboratory values (HbA1c, LDL cholesterol, PT-INR); patients' quality of life; utilisation frequency of healthcare services (hospital admissions, ED visits); economic outcomes.

Critical outcomes

Eleven trials reported medication adherence, assessed using self-reported tools (e.g. MMAS-8, MARS), pharmacy refill data, or

pill counts. Telepharmacy interventions may improve medication adherence compared to usual care (low-certainty evidence). Effects remained consistent across different intervention types and delivery methods. One additional study not included in the meta-analysis also reported improved adherence in the intervention group (Odegard 2012). These findings suggest that telepharmacy may improve adherence, although we downgraded the certainty of the evidence due to risk of bias, heterogeneity, and concerns about measurement validity.

Five trials assessed patients' satisfaction. Based on very-low certainty evidence from three RTCs, the evidence is very uncertain about the effect of telepharmacy interventions on patients' satisfaction. Two additional studies also favoured telepharmacy descriptively (Salmany 2018; Shdaifat 2022). Instruments varied widely across studies, limiting comparability and synthesis.

Six trials reported DRPs. One study aimed to detect DRPs and found a higher number of problems identified in the intervention group (Ibrahim 2022). Other studies aimed to reduce adverse events, such as bleeding or hypoglycaemia, with results mostly showing little to no difference between groups, although the direction of effect generally favoured the intervention. One study did not report group-level data (Salmany 2018), so could not contribute to comparative analyses. Measurement tools and outcome definitions differed, precluding meta-analysis. Overall, the findings reflected differences in outcome constructs across studies, with one study

addressing detection and four studies addressing prevention of DRPs. The certainty of the evidence was low.

Important outcomes

Two studies evaluated the worsening of NCDs. These studies reported clinically important deterioration events in diabetes and asthma, including diabetic ketoacidosis and acute exacerbations. However, the evidence was limited in quantity and clinically heterogeneous, and we reported the findings narratively. As a result, conclusions regarding the effect of telepharmacy on preventing clinical deterioration remain uncertain.

We disaggregated clinical measurements into five categories due to the broad definition used in the Cochrane protocol: asthma control, SBP, DBP, pain control, and thromboembolic events.

1. Two trials reported asthma control, but there was no clear difference between telepharmacy and usual care. An additional study reported ACT scores only in graphical form (Young 2012), which we could extract for meta-analysis.
2. Five trials assessed SBP and DBP. Based on low-certainty evidence, telepharmacy interventions may reduce SBP or DBP compared to usual care.
3. Pain control and thromboembolic events were assessed in two separate studies with distinct outcome measures. One study reported greater pain relief in the telepharmacy group; the other reported fewer thromboembolic events in the telepharmacy group.

We grouped laboratory values into HbA1c, LDL cholesterol, and PT-INR.

1. Five trials reported HbA1c. Based on moderate-certainty evidence, telepharmacy interventions probably have little or no effect on HbA1c compared to usual care. We excluded two additional studies from the meta-analysis—one due to inconsistent data presentation, and the other due to a categorical outcome format—but both showed trends favouring the telepharmacy group.
2. Two trials reported LDL cholesterol. Evidence suggested no clear difference between telepharmacy and usual care.
3. One trial assessed PT-INR control, which found a greater proportion of time in the therapeutic range in the telepharmacy group (45.6%) compared with the control group (24.1%; $P = 0.005$).

Three trials assessed patients' quality of life using different instruments to assess quality of life (e.g. EQ-5D-5L, PAQLQ). Results were mixed: one study reported an improvement, one reported similar levels between groups, and another reported within-group improvements without between-group analysis. Due to the small number of studies and heterogeneity of instruments and populations, findings remain uncertain.

We divided utilisation frequency of healthcare services into hospital admissions and ED visits.

1. Two trials reported unplanned hospital admissions. Evidence suggested no clear difference between telepharmacy and usual care. One additional study, which reported hospitalisation frequency as a continuous variable and was not included in the

meta-analysis, found fewer hospitalisations in the telepharmacy group.

2. Two trials reported ED visits with inconsistent findings: one found fewer visits in the intervention group, whereas the other found no difference.

Two trials reported economic outcomes narratively. One study from the USA found a reduction in total medical costs in the telepharmacy group, with mixed effects across cost categories: higher costs for hypertension medications and lower costs for lipid-lowering drugs. Another study from Jordan reported lower private medication costs in the intervention group (13.16 JD versus 26.74 JD; 1 JD = 1.4 USD).

No studies reported data on mortality rate, adverse events attributable to telepharmacy, or the satisfaction of other healthcare providers, limiting conclusions about potential harms.

Limitations of the evidence included in the review

Certainty of the evidence

The overall certainty of the evidence varied across outcomes. We assessed the certainty of evidence as moderate for laboratory values (HbA1c); low for medication adherence, DRPs, and clinical measurements (SBP and DBP); and very low for patient satisfaction. Often we downgraded the certainty of the evidence due to issues such as selective reporting, lack of pre-specified analysis plans, differences in outcome definitions, and heterogeneity in measurement tools. For outcomes not included in the summary of findings table (e.g. worsening of NCDs, healthcare utilisation frequency, economic outcomes), we did not perform formal GRADE assessments because the available evidence did not allow reliable effect estimation or application of GRADE. This was due to substantial heterogeneity in outcome definitions, reporting formats, and data completeness, with the limited number of studies further limiting interpretation. These limitations indicate important constraints on the overall certainty of the evidence presented in this Cochrane review.

Study design and conduct

Many included trials had important limitations in study design and conduct. Pre-registration was often absent, and prespecified analysis plans were rarely reported. For medication adherence, only one of 11 trials was at low risk of bias; we rated others as high risk due to deviations from intended interventions (Domain 2), reliance on self-report (Domain 4), and selective outcome reporting (Domain 5). For patient satisfaction, issues included poor allocation of concealment, lack of blinding, inconsistent timing and measurement tools. For DRPs, intervention intent varied. One trial aimed to detect DRPs (Ibrahim 2022), while others focused on preventing adverse events, limiting direct comparability with others aiming to reduce them. All three asthma control studies were at high overall risk of bias, with common concerns including intervention deviations, incomplete data, and missing pre-registration (Kosse 2019; Shdaifat 2022; Young 2012). All five SBP and DBP trials were individually randomised, moderately sized, and rated with some concerns, particularly for Domains 2 and 5 (Bosworth 2018; Ibrahim 2022; Margolis 2013; Mozu 2023; Scala 2018). For HbA1c, we also noted limitations in study design. For example, Gautier 2021 was a cluster-RCT that did not account for clustering in the analysis. Subgroup analyses yielded no clear effects. These methodological flaws increased bias risk and limited

interpretability. However, with increasing expectations for pre-registration and protocol transparency in clinical trials, future research may overcome some of these limitations.

Outcome reporting

Limitations in outcome reporting also contributed to challenges in interpreting and synthesising the evidence. For medication adherence, the included trials used a range of measurement tools (e.g. MMAS-8, MPR, PDC). Even with the use of SMDs, full comparability was not guaranteed. One study used a non-validated, study-specific scale (Hoda 2023). In the assessment of patient satisfaction, trials used different instruments, such as VAS and TSQ. We excluded two studies from meta-analysis due to missing data from the control group or incompatible scales (Salmany 2018; Shdaifat 2022). For DRPs, we noted differences in outcome definitions. One intervention (Ibrahim 2022) aimed to detect DRPs, while four interventions (Gonzales 2021; Hussain 2024; Powers 1983; Sudas Na Ayutthaya 2018) were designed to reduce adverse events, making it difficult to interpret pooled findings consistently. Among trials assessing SBP and DBP, limited detail was provided on the timing and methods of measurement (e.g. whether blood pressure was assessed during clinic visits or at home, type of automated devices used, number of measurements), raising concerns about data consistency. In HbA1c reporting, inconsistencies in values in the article and formats were apparent. One study reported dichotomised data only (Ihekoronye 2024). These factors contributed to difficulties in synthesising and interpreting the results across studies.

Generalisability of the findings

The generalisability of the findings may be limited. Most studies were conducted in HICs (e.g. 7/11 on medication adherence), and populations were primarily restricted to specific chronic conditions, such as diabetes, asthma, and hypertension. Variability in age, gender, and settings also reduces external validity. For patient satisfaction, a few trials included children, but their number was small. Telepharmacy delivery modes and intervention durations varied widely. HbA1c studies included diverse age groups, but as a diabetes-specific marker, its applicability to other conditions remains limited.

Publication bias

We were unable to rule out concerns about publication bias and the potential influence of unpublished data. Due to the limited number of included trials, formal assessment of publication bias was only possible for the outcome of medication adherence. For this outcome, a funnel plot based on 10 studies appeared approximately symmetrical, suggesting a low likelihood of publication bias. However, for all other outcomes, the number of studies was insufficient to allow such evaluation, and the risk of publication bias remains uncertain.

Limitations of the review processes

This systematic review process had several limitations. First, although we conducted comprehensive searches across multiple databases and grey literature sources without language restrictions, some databases (e.g. OpenGrey) were no longer active. Our attempts to contact study authors yielded limited responses, which may have resulted in missing or unclear data.

Second, several studies had missing or unclear outcome data (e.g. SDs, unit of analysis details, sample sizes). Where necessary, we calculated derived estimates in accordance with guidance from Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions*, or excluded studies from meta-analysis when we could not retrieve data [79].

Third, meta-analysis was not feasible for some outcomes due to substantial heterogeneity in outcome definitions, measurement tools, or intervention aims (e.g. DRPs, economic outcomes), requiring narrative synthesis. Additionally, the number of studies was fewer than 10 for most outcomes, precluding meaningful assessment of publication bias. For the single outcome with more than 10 studies, we generated a funnel plot in RevMan and assessed it visually.

Finally, interpretation of findings inevitably involved some subjective judgement. However, our multidisciplinary review team, including professionals from pharmacy, medicine, nursing, and biostatistics from across several countries, provided a balanced perspective that may have mitigated individual biases. In future review updates, continued collaboration and methodological rigour will be important for enhancing review reliability and transparency, particularly as several ongoing trials may provide additional evidence.

Agreements and disagreements with other studies or reviews

Medication adherence

Several prior systematic and primary studies have examined telepharmacy or related remote pharmacy services across various chronic conditions. For medication adherence, Niznik 2018 synthesised heterogeneous interventions and populations, including people with HIV and those in smoking cessation programmes, and did not conduct a meta-analysis because of variations in measurement tools and small numbers of studies [50]. Since that review, at least 15 additional randomised trials meeting our PICO criteria have been published. Our review, focusing exclusively on ambulatory care patients with NCDs and pharmacist-led interventions, included 10 randomised trials in a meta-analysis. We found that telepharmacy interventions may improve medication adherence compared to usual care, based on low-certainty evidence. Both reviews covered multinational studies without age restrictions. Differences between our review and Niznik 2018 include provider type (pharmacist-only versus multidisciplinary) and intervention formats (telephone only versus telephone plus mobile apps and other digital tools). Despite these differences, both support the potential benefit of telepharmacy for improving adherence.

Patients' satisfaction

For patients' satisfaction, we selected Allen 2022 for comparison because it was a non-randomised study in dialysis patients (an NCD population) that compared telephone-based pharmacist follow-up with usual care, reporting higher satisfaction in the telepharmacy group [149]. Our Cochrane review included five RCTs, and we found the evidence is very uncertain about the effect of telepharmacy interventions on patients' satisfaction (very-low certainty evidence). Both our review and Allen 2022 suggest generally positive perceptions, but methodological differences

and non-validated measurement tools may account for more favourable findings in the observational study.

DRPs

Regarding DRPs, we found no previous systematic reviews with the same scope as ours, which focused on pharmacist-led telepharmacy for outpatients with NCDs and addressed both proactive detection of medication errors and prevention of adverse events. We selected comparable studies and meta-analyses in populations and interventions most closely aligned with our PICO (e.g. similar conditions, telepharmacy-based monitoring, and relevant outcome definitions). For detection, one RCT in our review (Ibrahim 2022), and an observational study of telemedicine-based DOAC monitoring [150], showed that telepharmacy facilitated identification of problems. For prevention, one RCT reported fewer severe hypoglycaemic events (Hussain 2024), consistent with a 2022 meta-analysis including RCTs and observational studies in similar populations (some multidisciplinary) that found reduced hypoglycaemia (RR 0.48, 95% CI 0.30 to 0.76) [64]. For anticoagulated patients, our findings on major bleeding were broadly in line with Tran 2021, a meta-analysis largely of non-randomised trials in warfarin-treated patients using varied telepharmacy formats, which reported reduced bleeding (RR 0.65) [37]. Differences between our findings and other studies may reflect variation in intervention components, aims, and measurement tools, and differences in outcome constructs (detection versus prevention of DRPs), thus limiting direct comparability.

Asthma control

For asthma control, we compared our findings with a quasi-experimental one-group pre-post study, which conducted pharmacist-led telepharmacy for asthma via a digital platform [151]. This study found an increase in the proportion of well-controlled asthma after 12 weeks. In our review, a meta-analysis of two RCTs showed no clear difference in asthma control between telepharmacy and usual care. One 12-week trial indicated a possible benefit, but results varied across studies. Differences in study design and outcome measurement (continuous ACT, C-ACT, or CARAT scores versus dichotomous GINA categories) may explain the variation in findings.

Clinical measurements

For blood pressure, we selected Baral 2023 for comparison because it was a recent meta-analysis of telemonitoring interventions for hypertension that included pharmacist-led components as well as other providers, making it partially comparable to our PICO [152]. Baral 2023 included five RCTs and one quasi-experimental study, and found reductions in SBP and DBP favouring intervention. In our Cochrane review, based on low-certainty evidence, telepharmacy may reduce both SBP and DBP compared with usual care. By contrast, Baral 2023 reported greater reductions, possibly reflecting differences in provider type, inclusion of non-randomised evidence, and the broader scope of telemonitoring interventions.

Laboratory measurements

For HbA1c, our review included five randomised trials in people with NCDs. Based on moderate-certainty evidence, telepharmacy interventions probably have little or no effect on HbA1c compared to usual care. We reported two additional trials narratively because their results were not in a format suitable for pooling; both favoured

telepharmacy. We compared our findings with a meta-analysis of patients with diabetes, a condition within the NCD group, that included both randomised and observational studies [64]. That review involved some interventions delivered by multidisciplinary teams, whereas ours focused solely on pharmacist-led care. Certain delivery features, such as multi-component telepharmacy formats, were common to both. The meta-analysis reported a larger and greater reduction in HbA1c (MD -1.26%, 95% CI -1.69 to -0.84; $I^2 = 98.7\%$), likely reflecting differences in study design, inclusion of non-randomised evidence, provider type, and baseline HbA1c levels. Both systematic reviews indicate potential benefit for glycaemic control, but variations in evidence sources and participant characteristics likely contributed to the larger effect and higher heterogeneity observed previously.

We expect several ongoing studies, identified during the screening process (see [Supplementary material 5](#)), will further inform the evidence base on telepharmacy interventions.

AUTHORS' CONCLUSIONS

Implications for practice

This Cochrane review suggests that telepharmacy interventions for patients with non-communicable diseases (NCDs) in ambulatory care settings may offer improvements over usual care in several outcomes.

In particular, a meta-analysis of 10 randomised trials suggested that telepharmacy interventions may improve medication adherence compared to usual care. Although we rated the certainty of the evidence as low, the findings indicate potential applicability in practice. Thresholds, such as 80% adherence, have been used in some of the included studies, particularly those using refill-based measures such as medication possession ratio (MPR) or proportion of days covered (PDC), while self-report measures such as the 8-item Morisky Medication Adherence Scale (MMAS-8) classify adherence into categorical levels rather than a single cut-off. In light of this variation in measurement approaches, the applicability of any single general threshold for clinically meaningful adherence may vary across conditions. Medication adherence is a dynamic behaviour that may change over time, and changes in adherence status are therefore typically interpreted within the clinical context of each disease and treatment. Telepharmacy interventions may reduce systolic blood pressure (SBP) and diastolic blood pressure (DBP) compared with usual care, although the certainty of this evidence was low. For glycated haemoglobin (HbA1c), based on moderate-certainty evidence, telepharmacy interventions probably have little or no effect on HbA1c compared to usual care. The evidence base for drug-related problems (DRPs) remains limited and of low certainty. Most included studies only reported adverse events descriptively, with few addressing early detection or prevention of DRPs in a way that informs clinical decision-making.

For other outcomes—such as patient satisfaction—the effects of telepharmacy interventions remain uncertain, as the certainty of the evidence was generally low or very low. Accordingly, further evidence is needed before broadly implementing telepharmacy interventions targeting these outcomes in practice.

Telepharmacy services can be delivered via various channels, including telephone, video calls, and mobile applications.

However, successful implementation may require consideration of technical infrastructure, operational workflows, and training of healthcare professionals. Potential barriers in routine practice include unstable internet access, concerns over privacy, and variable patient digital literacy. These challenges may be particularly relevant when telepharmacy is used for complex goals, such as managing DRPs or multiple health-related outcomes. In such cases, careful contextual adaptation and phased implementation may be necessary.

Equity-related implications for practice

Most included studies were conducted in high-income countries (HICs), with none representing low-income settings. Given that telepharmacy interventions require stable digital infrastructure and reliable connectivity, implementation in resource-constrained environments may present substantial practical challenges. In settings with limited technology access, telepharmacy could potentially exacerbate disparities in healthcare access and medication continuity. When considering the adoption of telepharmacy, stakeholders may need to carefully assess the feasibility and infrastructure context of their specific settings.

Implications for research

The certainty of the evidence for medication adherence was low, limiting confidence in the findings. Future research should prioritise well-designed randomised trials with adequate sample sizes, standardised adherence measures, and longer follow-up to clarify the sustainability and clinical relevance of observed effects.

In particular, there was considerable variation in how the included trials assessed adherence—ranging from self-reported questionnaires to pharmacy claims and dispensing records. Future research should incorporate validated and standardised outcome measures to assess medication adherence. It is also important to clearly define the telepharmacy interventions (e.g. frequency of support; delivery mode; intervention components: adherence support, behavioural counselling; target disease; and patient characteristics) and to consider flexibility in adapting these types to individual or contextual needs. While this review explored some of these differences through subgroup analyses (e.g. delivery modes, multi-component versus single-component interventions), more robust data is needed to evaluate the effectiveness and feasibility of specific intervention components. This would help inform the optimal design and implementation of telepharmacy strategies tailored to different healthcare systems and settings.

In addition, the evidence regarding early detection and prevention of adverse drug events or serious side effects was limited. Future studies should adopt rigorous designs that enable evaluation of the clinical contribution of pharmacists, including clear specification of outcomes and, where feasible, blinded outcome assessment.

Furthermore, we noted limitations in the duration of follow-up. In the included trials, the maximum follow-up period was 12 months, which may not be sufficient to assess the long-term sustainability of behaviour change or medication support for chronic conditions such as NCDs. Longer follow-up periods will be necessary to determine the lasting impact of telepharmacy interventions.

Eight ongoing trials identified in this review will provide further evidence (see [Supplementary material 5](#)), particularly in asthma,

diabetes, kidney disease, and cancer populations. Future research should build on these trials and avoid unnecessary duplication.

Future research should also consider greater involvement of consumers and patients in the design and evaluation of telepharmacy interventions. It should examine the role of telepharmacy in conjunction with broader health system components, such as remuneration models, financing and resource allocation, and governance arrangements, to support effective and sustainable implementation.

Equity-related implications for research

None of the included studies reported outcomes stratified by PROGRESS-Plus factors, such as gender, education, or socioeconomic status [87]. We did not specify these variables in the published Cochrane protocol [53]. While we conducted a subgroup analysis based on country income level (as classified by the World Bank), future research should incorporate more granular equity-relevant factors (e.g. residence, gender, education) to better understand whether telepharmacy reduces or exacerbates health disparities. Further research is also needed to assess feasibility and access in rural or underserved settings.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015136.pub2](https://doi.org/10.1002/14651858.CD015136.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Summary of included studies: participant and study characteristics

Supplementary material 10 Summary of telepharmacy intervention characteristics across included studies

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article.

1. Sign-off Editor (final editorial decision): Sascha Köpke, University of Cologne
2. Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Sue Marcus, Cochrane Central Editorial Service
3. Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported the editorial team): Cynthia Stafford, Cochrane Central Editorial Service
4. Copy Editor (copy editing and production): Deirdre Walshe, Cochrane Central Production Service
5. Peer reviewers (provided comments and recommended an editorial decision):
 - a. Ghada A Bawazeer (PharmD, BCPS, BCMTM), King Saud University, Riyadh, Saudi Arabia (clinical/content review)
 - b. Jack Baldikoski (patient and public review)
 - c. Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review)
 - d. Jo Platt, Central Editorial Information Specialist (search review)
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Contributions of authors

1. Hideki Sugita (HS), Miki Takenaka Sato (MTS), Erika Ota (EO), and Takeshi Hasegawa (TH) conceived the review.
2. HS, MTS, and TH designed the review.

3. HS, MTS, TH, Shunsuke Toyoda (ST), Hisashi Noma (HN), and EO coordinated the review.
4. HS, MTS, Noyuri Yamaji (NY), Masayuki Maeda (MM), Takenori Ichimura (TI), and Tomiko Sunaga (TS) selected studies for inclusion.
5. HS, MTS, TI, TS, MM, NY, Etsuko Nishimura (EN), Phyu Phyu Tun (PPT), and Gessica Augustin (GA) performed data extraction.
6. HS, MTS, TI, TS, NY, EN, PPT, and GA performed the risk of bias assessments.
7. HS and NY assessed the certainty of the evidence (GRADE).
8. HS, MTS, NY, and HN performed data entry and management.
9. HS, NY, and HN analysed the data.
10. HS, MTS, TH, EO, TI, TS, MM, NY, ST, and HN wrote the review.
11. TH, HN, EO, and YN provided general advice on the review.
12. HS, MTS, NY, MM, TI, TS, ST, EN, PPT, GA, HN, EO, and TH read and approved the final version prior to publication.

Declarations of interest

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History

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Registration and protocol

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Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library: full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies (included ([Supplementary material 2](#)), excluded at the full text screen ([Supplementary material 3](#)), awaiting classification ([Supplementary material 4](#)), or ongoing ([Supplementary material 5](#))) in the final review; study data, including study information, study arms, and study results or test data ([Supplementary material 8](#)); consensus risk of bias assessments ([Supplementary material 6](#)); and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 7](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. We developed the summary of findings tables using GRADEpro GDT. No additional code or external data sets were used. We assessed the risk of bias using the Cochrane RoB 2 tool, and additional supporting materials (e.g. RoB 2 Excel sheets) are available from the Open Science Framework (OSF; DOI: 10.17605/OSF.IO/KBNZ3).

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ADDITIONAL TABLES

Table 1. Overview of synthesis and included studies

Outcome	Study ID	Design	Population/condition	Setting	Country	Intervention	Comparator
Critical outcomes							
Medication adherence	Meta-analysis (quantitative synthesis): Alsabbagh 2012; Gautier 2021; Hoda 2023; Ibrahim 2022; Ihekoronye 2024; Kosse 2019; Lauffenburger 2019; Margolis 2013; Mozu 2023; Sarayani 2018 Narrative synthesis: Odegard 2012	7 individual-ly RCTs and 4 cluster-RCTs	Asthma; cardiovascular diseases; hypertension; type 2 diabetes mellitus	Clinic; health insurer; hospital; pharmacy	Canada; France; Ghana; India; Iran; Netherlands; Nigeria; UAE; USA	Smartphone app; SMS; telephone; video calls	Usual care
Patients' satisfaction	Meta-analysis (quantitative synthesis): Bynum 2001; Margolis 2013; Powers 1983 Narrative synthesis: Salmany 2018; Shdaifat 2022	5 individual-ly RCTs and 1 cluster-RCT	Asthma; cancer; hypertension	Clinic; hospital	Jordan; USA	Smartphone; telephone; video conferencing	
DRPs	Narrative synthesis: Gonzales 2021; Hussain 2024; Ibrahim 2022; Powers 1983; Salmany 2018; Sudas Na Ayut-thaya 2018	6 individually RCTs	Cancer; cardiovascular disease; hypertension; kidney failure (transplant); type 1 diabetes mellitus	Clinic; hospital; pharmacy	Jordan; Pakistan; Thailand; UAE; USA	Mobile app; smartphone; SMS; telephone; video calls	
Important outcomes							
Worsening of NCDs	Meta-analysis (quantitative synthesis): Hussain 2024; Shdaifat 2022	2 individually RCTs	Asthma; type 1 diabetes mellitus	Hospital	Jordan; Pakistan	Mobile app; smartphone	Usual care
Clinical measurements (asthma control)	Meta-analysis (quantitative synthesis): Kosse 2019; Shdaifat 2022 Narrative synthesis: Young 2012	2 individual-ly RCTs and 1 cluster-RCT	Asthma	Hospital; pharmacy	Jordan; The Netherlands; USA	Smartphone app; telephone	
Clinical measurements (SBP)	Meta-analysis (quantitative synthesis): Bosworth 2018; Ibrahim 2022; Margolis 2013; Mozu 2023; Scala 2018	4 individual-ly RCTs and 1 cluster-RCT	Hypertension; hypercholesterolaemia	Clinic; hospital; pharmacy	Ghana; UAE; USA	SMS; telephone; video calls	

Table 1. Overview of synthesis and included studies (Continued)

Clinical measurements (DBP)	Meta-analysis (quantitative synthesis): Bosworth 2018; Ibrahim 2022; Margolis 2013; Mozu 2023; Scala 2018	4 individually RCTs and 1 cluster-RCT	Hypertension; hypercholesterolaemia	Clinic; hospital; pharmacy	Ghana; UAE; USA	SMS; telephone; video calls
Laboratory values (HbA1c)	Meta-analysis (quantitative synthesis): Bosworth 2018; Gautier 2021; Hoda 2023; Lauffenburger 2019; Sarayani 2018 Narrative synthesis: Hussain 2024; Ihekoronye 2024	6 individually RCTs and 1 cluster-RCT	Type 1 and 2 diabetes mellitus; hypercholesterolaemia; hypertension	Clinic; health insurer; hospital; pharmacy	France; India; Iran; Nigeria; Pakistan; USA	Mobile app; SMS; telephone
Laboratory values (LDL cholesterol)	Meta-analysis (quantitative synthesis): Bosworth 2018; Sarayani 2018; Sudas Na Ayutthaya 2018	3 individually RCTs	Cardiovascular disease; type 2 diabetes mellitus; hypercholesterolaemia; hypertension	Clinic; hospital; pharmacy	Iran; Thailand; USA	Telephone
Patients' quality of life	Narrative synthesis ^a : Hoda 2023; Kosse 2019; Shdaifat 2022	2 individually RCTs and 1 cluster-RCT	Asthma; type 2 diabetes mellitus	Hospital; pharmacy	India; Jordan; The Netherlands	Smartphone app; SMS; telephone
Utilisation frequency of healthcare services (hospital admissions)	Meta-analysis (quantitative synthesis): Salmany 2018; Sudas Na Ayutthaya 2018 Narrative synthesis: Gonzales 2021	3 individually RCTs	Cancer; cardiovascular disease; kidney failure (transplant)	Clinic, hospital	Jordan; Thailand; USA	Smartphone; telephone
Utilisation frequency of healthcare services (ED visits)	Narrative synthesis: Salmany 2018; Shdaifat 2022	2 individually RCTs	Asthma; cancer	Hospital	Jordan	Telephone
Economic outcomes	Narrative synthesis: Margolis 2013; Shdaifat 2022	1 individually RCT and 1 cluster-RCT	Asthma; hypertension	Clinic	USA	Telephone

Abbreviations: DBP: diastolic blood pressure; DRPs: drug-related problems; ED: emergency department; HbA1c: glycated haemoglobin; LDL: low-density lipoprotein; NCDs: non-communicable diseases; RCT: randomised controlled trial; SBP: systolic blood pressure; SMS: short message service; UAE: United Arab Emirates.

^aDescribed narratively (including outcomes where we attempted but did not report a meta-analysis, due to heterogeneity or insufficient number of included studies).